

ProxyVis—A Proxy for Nighttime Visible Imagery Applicable to Geostationary Satellite Observations

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ABSTRACT: Visible satellite imagery is widely used by operational weather forecast centers for tropical and extratropical cyclone analysis and marine forecasting. The absence of visible imagery at night can significantly degrade forecast capabilities, such as determining tropical cyclone center locations or tracking warm-topped convective clusters. This paper documents ProxyVis imagery, an infrared-based proxy for daytime visible imagery developed to address the lack of visible satellite imagery at night and the limitations of existing nighttime visible options. ProxyVis was trained on the VIIRS day/night band imagery at times close to the full moon using VIIRS IR channels with closely matching *GOES-16/17/18*, *Himawari-8/9*, and *Meteosat-9/10/11* channels. The final operational product applies the ProxyVis algorithms to geostationary satellite data and combines daytime visible and nighttime ProxyVis data to create full-disk animated GeoProxyVis imagery. The simple versions of the ProxyVis algorithm enable its generation from earlier GOES and Meteosat satellite imagery. ProxyVis offers significant improvement over existing operational products for tracking nighttime oceanic low-level clouds. Further, it is qualitatively similar to visible imagery for a wide range of backgrounds and synoptic conditions and phenomena, enabling forecasters to use it without special training. ProxyVis was first introduced to National Hurricane Center (NHC) operations in 2018 and was found to be extremely useful by forecasters becoming part of their standard operational satellite product suite in 2019. Currently, ProxyVis implemented for *GOES-16/18*, *Himawari-9*, and *Meteosat-9/10/11* is being used in operational settings and evaluated for transition to operations at multiple NWS offices and the Joint Typhoon Warning Center.

SIGNIFICANCE STATEMENT: This paper describes ProxyVis imagery, a new method for combining infrared channels to qualitatively mimic daytime visible imagery at nighttime. ProxyVis demonstrates that a simple linear regression can combine just a few commonly available infrared channels to develop a nighttime proxy for visible imagery that significantly improves a forecaster's ability to track low-level oceanic clouds and circulation features at night, works for all current geostationary satellites, and is useful across a wide range of backgrounds and meteorological scenarios. Animated ProxyVis geostationary imagery has been operational at the National Hurricane Center since 2019 and is also currently being transitioned to operations at other NWS offices and the Joint Typhoon Warning Center.

KEYWORDS: Clouds; Tropical cyclones; Algorithms; Remote sensing; Satellite observations; Statistical techniques

1. Introduction

Visible light sensors (0.4–0.7 μm) were the first to fly on meteorological satellites and traditionally had the highest spatial resolution (Kidder and Vonder Haar 1995). Visible imagery is

widely used for cloud feature detection and is particularly useful for the detection and tracking of oceanic low-level clouds, which are challenging to detect using IR channels (3.9–14.0 μm) because they have BTs (all acronyms are provided in Table 1) similar to the underlying surface.

In the tropics and oceanic regions, satellite data are often the only source of information on cloud location and distribution, and visible imagery is often the only observation of the near-surface atmospheric motion, which makes it one of the primary tools used at operational forecasting centers (e.g., NHC, CPHC, JTWC, OPC, and WPC) for TC and ETC location analysis and intensity estimates, and oceanic cloud monitoring. Static visible imagery supports identification of synoptic and mesoscale features (Dvorak and Smigielski 1990a,b), and animated visible geostationary satellite imagery enables inference of atmospheric motion by both

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TABLE 1. List of acronyms, abbreviations, and ProxyVis definitions.

ABI	Advanced Baseline Imager
AHI	Advanced Himawari Imager
AMI	Advanced Meteorological Imager
ATCF	Automated TC Forecast System
ATCF al	ATCF Atlantic basin
ATCF ep	ATCF eastern Pacific basin
ATCF wp	ATCF western Pacific basin
ATCF io	ATCF Indian Ocean basin
BT	Brightness temperature
CGAN	Conditional generative adversarial network
CPHC	Central Pacific Hurricane Center
DNB	Day–night band
ETC	Extratropical cyclone
GeoIPS	Geolocated Information Processing System
<i>Geo-KOMPSAT-2A</i>	<i>Geostationary Korea Multi-Purpose Satellite-2A</i>
HSU	Hurricane Specialist Unit
ISATSS	Integrated Dissemination Program (IDP) Satellite Subsystem
IWTC-10	WMO Tenth International Workshop on TCs
JPSS	Joint Polar Satellite Systems
JTWC	Joint Typhoon Warning Center
LLCC	Low-level circulation center
ML	Machine learning
MSG	Meteosat Second Generation
MTG	Meteosat Third Generation
NHC	National Hurricane Center
NOAA CLASS	NOAA Comprehensive Large Array-data Stewardship System
NWS WFO	NWS Weather Forecast Office
OLS	Operational Linescan System
OPC	Ocean Prediction Center
PG	Proving Ground
SEVIRI	Spinning Enhanced Visible and Infrared Imager
SLIDER	Satellite Loop Interactive Data Explorer in Real-time
SSIM	Structural similarity index measure
SSO satellites	Sun-synchronous orbit satellites
SWD	Split window difference
TC	Tropical cyclone
TAFB	Tropical Analysis and Forecast Branch
WPC	Weather Prediction Center
ProxyVis	A common name used to describe any of the four ProxyVis algorithms described in section 2e
GeoProxyVis	The full disk geostationary product that uses ProxyVis for the nighttime part and the adjusted visible for the daytime part

manual and automated methods ([Velden et al. 1997, 1998](#); [Holmlund et al. 2001](#); [Velden et al. 2005](#)). One of the most important uses of visible imagery by TC forecast centers is tracking oceanic low-level clouds and locating TC LLCCs. However, visible imagery is not available at night, leading to significant degradation of low-level cloud detection at nighttime.

RGB products (e.g., [Lensky and Rosenfeld 2008](#)) can help identify low-level oceanic clouds. One noteworthy product is the Nighttime Microphysics RGB that differentiates low-level (liquid water) clouds from the background surface. However, similar to other RGBs, Nighttime Microphysics requires color displays, which makes its appearance different from the typical grayscale visible imagery, and thus a special grayscale product that simulates visible imagery during nighttime, captures oceanic low-level clouds, and

can be seamlessly combined with grayscale visible imagery for long animation containing day and nighttime scenes is desirable.

Two approaches have been used in operations as alternatives to visible imagery at night: 1) low-light sensors that provide visible-like imagery in near-full moonlight conditions, and 2) methods for extracting visible-like information from IR channels.

There are two operational low-light sensors, the DMSP OLS ([Dickinson et al. 1974](#)) and the JPSS VIIRS DNB ([Lee et al. 2006](#); [Miller et al. 2013](#)). Both fly on-board SSO satellites that have two major disadvantages at low latitudes: limited temporal coverage (twice per day) and up to a 2-h data latency if the desired target is not available via direct broadcast. In addition, the image quality for both sensors strongly depends on the moon phase. The DNB, which offers higher

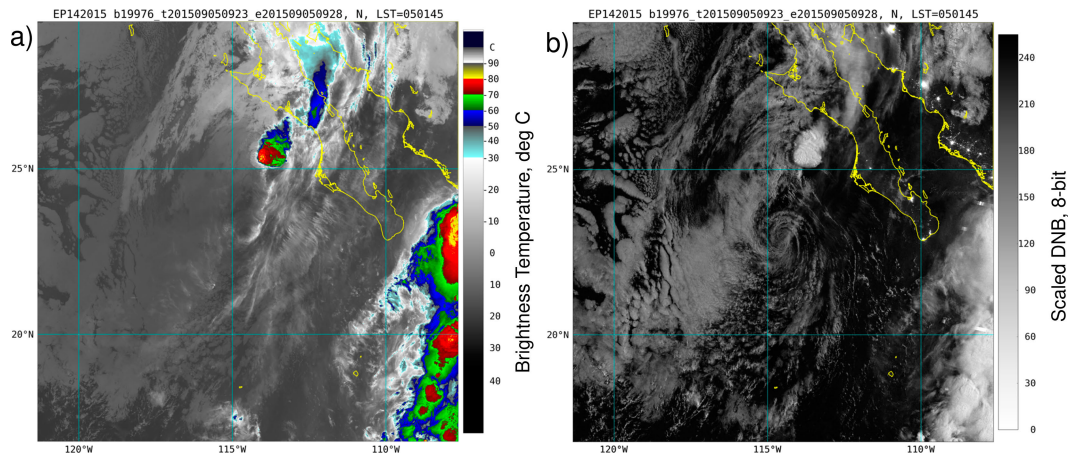


FIG. 1. The LLCC is difficult to see in (a) IR imagery (shown is VIIRS channel I05, $11.45\ \mu\text{m}$), but is easily identified in (b) the VIIRS scaled DNB (see [appendix B](#)). The figure is for the eastern Pacific Tropical Depression Kevin (2015) at 0928 UTC 5 Sep 2015. The moon is 49% full.

sensitivity¹ than the OLS ([Lee et al. 2006](#); [Miller et al. 2013](#); [Hillger et al. 2016](#)), is useful for nighttime meteorological observations in the tropics during the approximately half period of the 29.3-day lunar cycle when the moon is above the horizon at the time of the satellite overpass. Thus, even though DNB can capture TC LLCCs not visible in the IR channels ([Fig. 1](#)), even with VIIRS flying on three JPSS satellites, the availability of DNB is so limited that it is rarely considered for operational use at NHC or JTWC. In contrast, current geostationary satellites provide continuous 10–15-min animated imagery over their nearly hemispheric field of view.

The second approach includes a hierarchy of methods that utilize IR channels to simulate visible reflectance at nighttime ranging from single channel (e.g., $3.9\ \mu\text{m}$) to multiple channel algorithms. These methods heavily utilize the shortwave IR $3.9\text{-}\mu\text{m}$ channel ([Velden et al. 2005](#)) that has been available on *GOES-4–7* sounders ([VAS 2019](#)) and *GOES-8–18* imagers. The most widely used single channel method stretches² (see [appendix A](#)) $3.9\text{-}\mu\text{m}$ imagery to highlight the slightly cooler clouds over warm ocean background, and is used, for example, by automated methods for estimating low-level winds ([Dunjon and Velden 2002](#); [Velden et al. 2005](#)). This $3.9\text{-}\mu\text{m}$ product is reminiscent of visible imagery, but its quality is significantly affected by the SST background to which it is tuned (typically 26° – 29°C SST for tropical applications), and it quickly degrades over cooler or warmer SSTs. The legacy 2-channel method (low-cloud and fog product, hereafter “fog product,” [Ellrod 1995](#)) uses the difference between 10.7- and $3.9\text{-}\mu\text{m}$ channels. Its major limitation is that in the presence of large amounts of water vapor (common in the

tropics), the signal from low-level clouds often is masked. Versions of stretched $3.9\text{-}\mu\text{m}$ and fog products have been operational at NHC since 1994, when the $3.9\text{-}\mu\text{m}$ channel became available on the *GOES-8* imager. [Figure 2](#) shows the shortcomings of these products compared to DNB.

The first multichannel method, the original version of ProxyVis imagery described in this paper, was developed by [Chirokova et al. \(2018\)](#). Since then, other multichannel ML methods have been developed ([Kim and Hong 2019](#); [Kim et al. 2019](#); [Harder et al. 2020](#); [Pasillas et al. 2023](#)). [Kim and Hong \(2019\)](#) and [Kim et al. \(2019\)](#) use CGAN, one of the most complex artificial neural network models. [Pasillas et al. \(2023\)](#) developed nighttime visible imagery using a feed-forward neural network and followed the approach by [Chirokova et al. \(2018\)](#) to train their algorithm on VIIRS and apply it to ABI channels. While these methods are promising, it might be challenging to interpret what the neural network learned. To the best of our knowledge, none of these multichannel methods have been implemented in operations, except for [Kim and Hong \(2019\)](#) and [Kim et al. \(2019\)](#) that has been operational at the Korean Meteorological Agency for *Geo-KOMPSAT-2A* since March 2021 ([Sohn 2021](#)).

This paper describes ProxyVis imagery, a nighttime visible imagery proxy constructed from IR channels to address the limitations of existing methods for providing operational visible-like imagery at night. [Section 2](#) discusses data selection, preprocessing, and methods used. [Section 3](#) discusses ProxyVis comparison with legacy products, application to *GOES-16* data, and meteorological applications. [Section 4](#) summarizes the use of ProxyVis imagery in NWS and JTWC operations, the application of the ProxyVis algorithm to all current geostationary satellites, and product strengths and limitations. [Section 5](#) provides a summary and outlines future work.

2. Data and methods

ProxyVis is the first statistical model that was trained on VIIRS DNB to reproduce visible data from IR channels and

¹ DNB and OLS sensitivity is used in terms of both lower nominal minimum detection radiance at higher signal-to-noise ratio, and significantly higher radiometric resolution.

² “Stretched” $3.9\text{-}\mu\text{m}$ imagery uses a significant portion of the color table for a narrow range of values, see [appendix A](#).

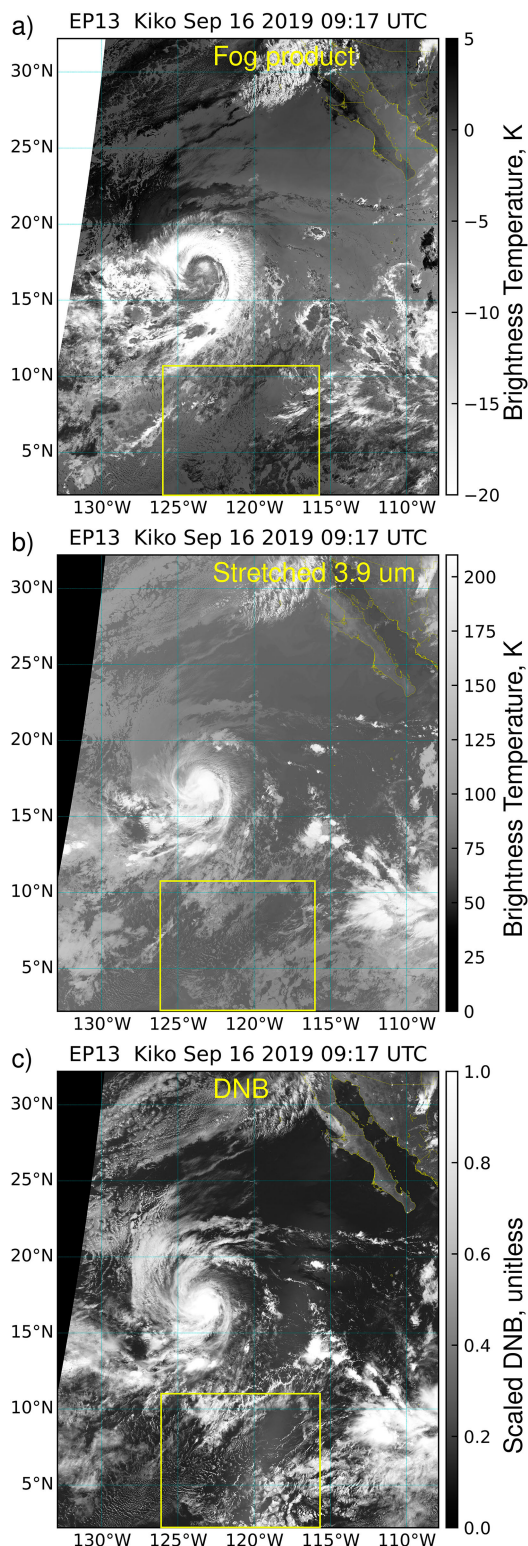


FIG. 2. Comparison of VIIRS data for (a) the fog product, $10.35 \mu\text{m}$ minus $3.9 \mu\text{m}$, (b) stretched $3.9\text{-}\mu\text{m}$, and (c) scaled DNB image for the eastern Pacific Major Hurricane Kiko (2019) at 0917 UTC 16 Sep 2019. The moon is 95% full. The values for the fog product for this scene range from -51 to 24 K, however, only the range of

apply that algorithm to geostationary data to create animated DNB-like imagery from geostationary data. In this section, *GOES-16* ABI data are used to describe the ProxyVis algorithm (except in section 2b where we need to take into account differences between ABI and SEVIRI channels); however, similar channels are also available on *GOES-17/18*, *Himawari-8/9*, *Geo-KOMPSAT-2A*, *Meteosat-9/10/11*, and *MTG*. The data processing and algorithm development consists of the following steps described in more detail in the subsections below:

- (i) selection of the statistical method (section 2a);
- (ii) selection of channels (section 2b);
- (iii) selection of training scenes and data preprocessing (section 2c);
- (iv) selection of predictors and derivation of coefficients (section 2d);
- (v) description of the final mathematical form of regression equations for four versions of ProxyVis (section 2e);
- (vi) comparison of the four versions of ProxyVis and suggested use for each version (section 2f);
- (vii) evaluation of the functional form of the algorithm (section 2g);
- (viii) independent verification (section 2h); and
- (ix) steps to combine daytime visible and nighttime ProxyVis imagery to create GeoProxyVis imagery, the final full-disk geostationary animated product (section 2i).

a. Selection of the statistical method

To develop the ProxyVis algorithm, we used multiple linear regression because it is one of the simplest available methods that is easy to train and interpret. Further, linear regression is usually more robust than nonlinear ML methods when running the model with data outside of the training set. Thus, our algorithm is more likely to be successful when trained on VIIRS data and applied to multiple geostationary satellite sensors using approximately matching channels, and forecasters can easily understand how the individual channels are being used. Using only VIIRS data for development ensures that the independent variables (IR channels) and the “truth” data (DNB) see objects at the same angle and time from the same platform, while assuming that VIIRS and ABI channels are similar to first order.

b. Selection of channels

We used as predictors the VIIRS IR channels that closely match channels available on all current geostationary satellites. In the first step, similar VIIRS, *GOES* ABI, *Himawari* AHI, and *MSG* SEVIRI channels were identified (Table 2). These include three longwave IR channels: VIIRS channels M14 ($8.55 \mu\text{m}$), M15 ($10.76 \mu\text{m}$), and M16 ($12.0 \mu\text{m}$)

BT differences from -20 to $+5$ K that contains most of the information is plotted. For this scene, there are 3.0% of pixels below -20 K and 0.18% of values above $+5$ K. The yellow rectangle highlights the area where low-level clouds are hard to distinguish in the fog product and stretched $3.9\text{-}\mu\text{m}$ image, but offers high contrast against the background in the DNB image.

TABLE 2. Visible (Vis, DNB), near-infrared (NIR), and infrared (IR) channels that are similar between *Suomi NPP* and *NOAA-20/21* VIIRS and GOES ABI, Himawari AHI, and MSG SEVIRI. Columns that show VIIRS values are highlighted in bold. The DNB provides dynamic range from day to night, while the ABI C02 and similar AHI and SEVIRI channels do not provide a nighttime visible capability. VIIRS “M” channels have nadir resolutions of 750 m, while VIIRS “I” channels have nadir resolutions of 375 m. VIIRS DNB channel has a resolution of 742 m across the swath. ABI and AHI IR channels have 2-km resolutions, ABI and AHI visible channels C02 and B03 have 0.5-km resolutions, and all SEVIRI channels listed in this table have 3-km resolutions.

Channel				Central wavelength (μm)				Wavelength range (μm)			
VIIRS	ABI	AHI	SEVIRI	VIIRS	ABI	AHI	SEVIRI	VIIRS	ABI	AHI	SEVIRI
M12	C07	B07	IR039	3.70	3.90	3.9	3.92	3.61–3.79	3.80–4.00	3.74–3.96	3.48–4.36
M13	C07	B07	IR039	4.05	3.90	3.9	3.92	3.97–4.13	3.80–4.00	3.74–3.96	3.48–4.36
I04	C07	B07	IR039	3.74	3.90	3.9	3.92	3.55–3.93	3.80–4.00	3.74–3.96	3.48–4.36
M14	C11	B11	IR087	8.55	8.50	8.6	8.7	8.40–8.70	8.30–8.70	8.44–8.76	8.3–9.1
M15	C13	B13	IR108	10.76	10.35	10.4	10.8	10.26–11.26	10.10–10.60	10.3–10.6	9.8–11.8
M16	C15	B15	IR120	12.01	12.30	12.3	12.0	11.54–12.49	11.80–12.80	12.2–12.5	11.0–13.0
DNB	C02	B03	VIS006	0.70	0.64	0.64	0.635	0.50–0.90	0.59–0.69	0.63–0.66	0.56–0.71

corresponding to ABI channels C11 (8.5 μm), C13 (10.35 μm), and C15 (12.3 μm), and to AHI and SEVIRI channels provided in Table 2. The selection of a shortwave VIIRS IR channel matching ABI channel C07 (3.9 μm) and similar AHI and SEVIRI channels was more complicated since VIIRS does not have a channel centered at 3.9 μm , but has three similar channels, M12 (3.7 μm), M13 (4.05 μm), and I04 (3.74 μm). Channel I04 was selected based on the comparison of the spectral response of these VIIRS channels (Fig. 4 from Hillger and Kopp 2019) with the ABI 3.9- μm channel (Fig. 1. From Schmit et al. 2017). The broader I04 band contains much of the ABI’s 3.9- μm band range and the responses to dry air and water vapor are nearly identical. Finally, VIIRS DNB was used as the dependent variable. The DNB center wavelength (0.70 μm) closely matches ABI “red” channel C02 (0.64 μm) that ProxyVis is trying to approximate, and which has been long used for visible satellite applications (Kidder and Vonder Haar 1995).

c. Selection of training scenes and preprocessing of VIIRS DNB data

In total 16 ProxyVis training scenes (Table 3) were selected from the CIRA TC-centric VIIRS database (currently 2012–23) that includes Sensor Data Record data collocated with TCs from the ATCF (Sampson and Schrader 2000) final best tracks. The VIIRS database provides ~ 10 min of data (~ 2000 km) around TC centers for all bands for each available VIIRS/TC overpass. The most important features that we wanted to enhance in the new product were the low-level oceanic clouds. Thus, the training scenes had to satisfy the following three criteria. First, a scene must occur within three days of a full moon to ensure well-lit scenes. Second, it should not contain any land (or land points should be filtered out) to ensure that background emissivity and radiance are more homogeneous. Third, a scene should be composed only of clouds and their surface backgrounds, and not contain significant additional signals such as lightning, moon glint, city lights, ship lights, etc. The scenes satisfying the above three criteria were manually selected from the CIRA archive of TC-centric VIIRS DNB imagery by subjectively evaluating each

available image within three days of the full moon. That reduced the number of available training scenes to less than 50 images. The final set of 15 TC training scenes was chosen subjectively by visual inspection of images to contain different cloud types [low, middle, and high level, as defined by WMO (2023)] over a wide range of SSTs, and TCs ranging from pre-TC disturbances and depressions to major hurricanes. One non-TC scene over the Gulf of Mexico was added to increase the number of developmental data points over very warm SST backgrounds.

Next, developmental DNB data were scaled to produce reflectance-like images. The DNB radiances span eight orders of magnitude, from $1.0 \times 10^{-10} \text{ W cm}^{-2} \text{ sr}^{-1}$ to at least $2.0 \times 10^2 \text{ W cm}^{-2} \text{ sr}^{-1}$ (Cao 2013), thus the radiances need to be scaled to approximate visible reflectance (Miller and Turner 2009; Seaman and Miller 2015). For this study, we used “gamma DNB” scaling (appendix B) developed at CIRA in 2014 to display TC-centric DNB imagery on RAMMB/CIRA (2019). The “gamma DNB” is normalized between 0 and 1, making it a convenient independent variable for ProxyVis training. Further, the developmental VIIRS data from I-, M-, and DNB channels were regridded to a common grid with 148 samples per degree latitude, which is representative of DNB resolution in the tropics. After regridding, the 16 training scenes (Table 3) contain a total of 123 616 263 pixels, with an average of 7 726 016 pixels per scene.

d. Selection of predictors and derivation of coefficients for multiple regression

The ProxyVis regression equations were derived in several steps. Initial testing was performed using multiple combinations of 2–3 scenes from Table 3 with different prevailing types of clouds (low clouds, deep convection, or cirrus), and different independent variable transformations. By visually comparing the images produced using IR channels with the corresponding DNB images, a combination of the VIIRS IR channels that yielded qualitatively satisfactory results was identified. This qualitative first pass at performance is based on the observation that common error metrics, such as RMSE, MAE, or SSIM might not correctly reflect the visual

TABLE 3. Details of VIIRS scenes used to develop ProxyVis imagery, including the ATCF storm ID (see Table 1), the storm name, the latest best track intensity, the times associated with DNB data collection, the percent of full moon, and the day of the closest full moon. Storm intensity is provided in knots (kt), and information about the moon phase was obtained from the U.S. Naval Observatory (2023).

Scene No.	Storm ID/name, latest best track intensity	Area bounds	Start time	End time	Percent full and closest full moon
1	al102015/Ida, 30 kt	15°N, 30°N, 55°W, 38°W	0425:185 UTC 26 Sep 2015	0430:589 UTC 26 Sep 2015	96%, 28 Sep
2	al102015/Ida, 30 kt	15°N, 30°N, 55°W, 40°W	0404:574 UTC 27 Sep 2015	0412:032 UTC 27 Sep 2015	99%, 28 Sep
3	al062016/Fiona, 30 kt	5°N, 20°N, 45°W, 25°W	0411:442 UTC 17 Aug 2016	0420:154 UTC 17 Aug 2016	99%, 18 Aug
4	al152016/Nicole, 60 kt	25°N, 50°N, 65°W, 35°W	0544:266 UTC 15 Oct 2016	0547:161 UTC 15 Oct 2016	99%, 16 Oct
5	al152016/Nicole, 75 kt	25°N, 50°N, 65°W, 35°W	0524:055 UTC 16 Oct 2016	0528:205 UTC 16 Oct 2016	Full, 16 Oct
6	al042017/TD04, 25 kt	5°N, 30°N, 60°W, 35°W	0433:090 UTC 7 Jul 2017	0435:586 UTC 7 Jul 2017	97%, 9 Jul
7	al042017/TD04, 25 kt	5°N, 30°N, 60°W, 35°W	0415:387 UTC 8 Jul 2017	0425:353 UTC 8 Jul 2017	99%, 9 Jul
8	al042017/TD04, 25 kt	5°N, 35°N, 65°W, 40°W	0555:135 UTC 8 Jul 2017	0600:539 UTC 8 Jul 2017	99%, 9 Jul
9	ep952017/Invest, 25 kt	0°, 25°N, 130°W, 95°W	0818:194 UTC 6 Jul 2017	0821:090 UTC 6 Jul 2017	93%, 9 Jul
10	wp042015/Maysak, 130 kt	5°N, 20°N, 130°E, 145°E	1657:332 UTC 1 Apr 2015	1704:390 UTC 1 Apr 2015	92%, 4 Apr
11	wp042015/Maysak, 105 kt	5°N, 20°N, 125°E, 140°E	1638:375 UTC 2 Apr 2015	1645:433 UTC 2 Apr 2015	96%, 4 Apr
12	wp052015/Haishenm 45 kt	5°N, 15°N, 145°E, 160°E	1600:479 UTC 4 Apr 2015	1607:519 UTC 4 Apr 2015	Full, 4 Apr
13	wp132015/Soudelor, 35 kt	5°N, 20°N, 150°E, 165°E	1545:365 UTC 31 Jul 2015	1552:405 UTC 31 Jul 2015	Full, 31 Jul
14	wp132015/Soudelor, 45 kt	5°N, 20°N, 145°E, 160°E	1526:408 UTC 1 Aug 2015	1533:466 UTC 1 Aug 2015	98%, 31 Jul
15	wp132015/Soudelor, 135 kt	10°N, 25°N, 135°E, 150°E	1628:242 UTC 3 Aug 2015	1636:554 UTC 3 Aug 2015	87%, 31 Jul
16	Gulf of Mexico/N/A, N/A	5°N, 35°N, 105°W, 55°W	0713:018 UTC 9 Jul 2017	0730:037 UTC 9 Jul 2017	Full, 9 Jul

quality of an image to human analysts (Dosselmann and Yang 2011). The selected channel combination is comprised of 1) channel I04 (3.74 μm), 2) channel difference [M15 (10.76 μm) – M16 (12.01 μm)], and 3) channel difference [I04 (3.74 μm) – M14 (8.55 μm)]. The best visual results were obtained by using nonlinear transforms of those three independent variables, as described in sections 2e and 2g.

After a preliminary form of the regression terms was determined, the exact form of independent variables was derived by minimizing the residuals, again using different combinations of 2–3 developmental scenes at a time. Finally, data from all 16 developmental scenes (>120 million points) were used to determine the regression coefficients. That was the maximum amount of data that could be processed using available computer memory (132 GB); however, given the number and the variety of training scenes and points and the independent verification results that had comparable R^2 values (section 2h), we do not think that adding more tropical scenes would improve the algorithm. Our training approach helped to ensure that the resulting images appear to the human eye like visible imagery and avoided some pitfalls of

automatic variable selection routines that could lead to overfitting, biases, and poor independent performance (Sainani 2013). Further, it was found that qualitatively the resulting imagery can be further improved by using two regression equations, one for low/warm clouds and a second for high/cold clouds. A downside to this approach is that the “boundary” between the two regressions can sometimes be seen in the ProxyVis imagery. However, the overall performance of the resulting algorithm is better compared to using a single regression for all clouds.

e. Regression equations for four versions of the ProxyVis algorithm

In this section, we describe four versions of the ProxyVis algorithm that enable its creation using the global constellation of geostationary satellites as well as its use in multispectral algorithms as a replacement for visible data. Table 4 summarizes the differences, similarities, and suggested use for each version. The comparison and the reasons for using four versions are further discussed in section 2f.

TABLE 4. The main differences and the suggested use for different versions of the ProxyVis algorithm. Column 2 shows the ABI channels used for each version of ProxyVis. Mapping of ABI channels to other satellites is shown in Table 2.

ProxyVis product version	Formulation	Best for	Limitations	Suggested satellite sensors for qualitative use	Suggested satellite sensors for quantitative use (i.e., as part of other algorithms, such as ML or RGBs)
PrVis—The main operational algorithm	Two regressions that use GOES ABI channels C07, C11, C13, and C15	Low-level clouds	Visible line between high and low clouds; decreased quality for midlevel clouds	1) GOES ABI 2) Himawari AHI 3) <i>Geo-KOMPSAT-2A</i> AMI 4) MSG SEVIRI	
PrVisA	Single regression that uses GOES ABI channels C07, C11, C13, and C15	All clouds	Decreased quality for low-level clouds		1) GOES ABI 2) Himawari AHI 3) <i>Geo-KOMPSAT-2A</i> AMI 4) MSG SEVIRI
PrVisSimple	Two regressions that use GOES ABI channel C07	Low-level clouds on satellites that do not have all channels for PrVis	Visible line between high and low clouds; decreased quality for midlevel clouds	1) MSG SEVIRI 2) GOES Imager (e.g., <i>GOES-13/15</i>) 3) <i>GOES-17</i> ABI (loop heat pipe) 4) Low-latency option for all satellites	
PrVisSimpleA	Single regression that uses GOES ABI channel C07	All clouds on satellites that do not have all channels for PrVis	Decreased quality for low-level clouds		1) MSG SEVIRI 2) GOES Imager (e.g., <i>GOES-13/15</i>) 3) <i>GOES-17</i> (loop heat pipe) 4) Low-latency option for all satellites

TABLE 5. Means and standard deviations for variables used in (2)–(9), where BT_{tr} is defined by (1), and C07, C11, C13, and C15 are BTs (K).

Variable	Mean	Std dev
Low clouds ($C07 > BT_{tr}$)		
Dependent variable (VIIRS DNB)	0.187	0.141
$ C13 - C15 ^{0.4}$	1.219	0.385
$\ln(C11 - C07)$	1.643	0.775
$C07^5$	2078.4×10^9	234.1×10^9
High clouds ($C07 \leq BT_{tr}$)		
Dependent variable (VIIRS DNB)	0.499	0.178
$ C13 - C15 ^{0.4}$	0.991	0.402
$\ln(C11 - C07)$	1.770	1.119
$C07^5$	1003.0×10^9	353.6×10^9
All clouds		
Dependent variable (VIIRS DNB)	0.266	0.203
$ C13 - C15 ^{0.4}$	1.161	0.402
$\ln(C11 - C07)$	1.675	0.877
$C07^5$	1805.1×10^9	540.3×10^9

1) MULTIPLE REGRESSION—MATHEMATICAL FORMULATION

The mathematical form of the multiple regression equations is written in terms of the ABI channels C07 (3.9 μm), C11 (8.4 μm), C13 (10.3 μm), and C15 (12.3 μm) instead of VIIRS channels because the primary operational use of the algorithm is with geostationary data. In the next five subsections, we discuss the regression equations for each of the four versions and then present the postprocessing steps that are applied in the same way to all four versions.³

(i) PrVis—Two regressions

The primary operational ProxyVis formulation, the *two-regression* version, combines two equations derived separately for “low/warm” and “high/cold” clouds. The boundary between “low” and “high” clouds is defined using an empirically selected threshold of

$$BT_{tr} = 273 \text{ K} \quad (1)$$

in ABI channel C07 BT. The equations for the “low/warm” clouds, PV_L , and “high/cold” clouds, PV_H , are shown in (2) and (3), where C07, C11, C13, and C15 are BTs (K), normalized by the sample mean and standard deviation (provided in Table 5), and the normalized coefficients are $X_{L1...3}$, and $X_{H1...3}$ (provided in Table 6):

$$PV_L = X_{L1}|C13 - C15|^{0.4} + X_{L2} \ln(|C11 - C07|) + X_{L3} C07^5, \quad (2)$$

$$PV_H = X_{H1}|C13 - C15|^{0.4} + X_{H2} \ln(|C11 - C07|) + X_{H3} C07^5. \quad (3)$$

³ ProxyVis has been presented at AMS conferences (e.g., Chirokova et al. 2018); however, the algorithm has not been previously documented in a peer-reviewed publication.

TABLE 6. Variance explained for the developmental sample and normalized coefficients used to estimate PV_L and PV_H (two left column) defined in (2) and (3) for the full PrVis, where the threshold, BT_{tr} , is defined by (1), and the same for PV_{SL} and PV_{SH} (two right columns) defined in (6) and (7) and used in PrVisSimple. C07, C11, C13, and C15 are BTs (K) for the corresponding ABI channels.

Main version		Simple version	
Low clouds ($C07 > BT_{tr}$), N data points = 92 205 242			
R^2	45.7%	R^2	41.7%
X_{L1}	−0.0622	—	—
X_{L2}	−0.1535	—	—
X_{L3}	−0.6036	X_{SH}	−0.6458
High clouds ($C07 \leq BT_{tr}$), N data points = 31 411 021			
R^2	39.6%	R^2	35.2%
X_{H1}	−0.1487	—	—
X_{H2}	−0.0902	—	—
X_{H3}	−0.5422	X_{SH}	−0.5930
All clouds estimated from low and high, N data points = 123 616 263			
R^2	68.7%	R^2	66.5%
All clouds, N data points = 123 616 263			
R^2	68.2%	R^2	65.7%
X_{A1}	−0.0682	—	—
X_{A2}	−0.1081	—	—
X_{A3}	−0.7871	X_{SA}	−0.8102

The equation for all clouds is obtained by combining (2) and (3) using BT_{tr} from (1):

$$PV = \begin{cases} PV_L, & \text{if } C07 \geq BT_{tr} \\ PV_H, & \text{if } C07 < BT_{tr} \end{cases}. \quad (4)$$

PV_L and PV_H should be unnormalized (converted back to dimensional quantities) after being estimated from (2) and (3). The corresponding values should be also unnormalized after being estimated from (5); (6) and (7); and (9).

(ii) PrVisA—Single regression

The regression equation for the version of ProxyVis that uses a *single regression* for clouds at all levels, is given by

$$PV_A = X_{A1}|C13 - C15|^{0.4} + X_{A2} \ln(|C11 - C07|) + X_{A3} C07^5, \quad (5)$$

where $X_{A1...3}$ are the normalized coefficients (Table 6), and the means and standard deviations for all terms are listed in Table 5.

(iii) PrVisSimple—Two regressions, single channel

PrVisSimple is a *single-channel* version of PrVis. Similar to (2)–(4), the equations for the low/warm clouds, PV_{SL} , and high/cold clouds, PV_{SH} , are shown in (6) and (7), where C07 is BT in kelvin, normalized by the sample mean and standard deviation (Table 5), and the normalized coefficients are X_{SL} and X_{SH} (Table 6):

$$PV_{SL} = X_{SL} C07^5, \quad (6)$$

$$PV_{SH} = X_{SH} C07^5. \quad (7)$$

The regression equation for all clouds is obtained by combining (6) and (7) using BT_{tr} from (1):

TABLE 7. Correlation matrix for regression terms used in Eqs. (2), (3), and (5).

	All clouds			Low clouds			High clouds		
	$C07^5$	$\ln(C11 - C07)$	$ C13 - C15 ^{0.4}$	$C07^5$	$\ln(C11 - C07)$	$ C13 - C15 ^{0.4}$	$C07^5$	$\ln(C11 - C07)$	$ C13 - C15 ^{0.4}$
$C07^5$	1.00	0.00	0.34	1.00	0.18	0.23	1.00	0.01	0.34
$\ln(C11 - C07)$		1.00	0.66		1.00	0.75		1.00	0.61
$ C13 - C15 ^{0.4}$			1.00			1.00			1.00

$$PV_S = \begin{cases} PV_{SL}, & \text{if } C07 \geq BT_{tr} \\ PV_{SH}, & \text{if } C07 < BT_{tr} \end{cases}. \quad (8)$$

(iv) *PrVisSimpleA—Single regression, single channel*

Finally, PrVisSimpleA is a *single-channel* version of PrVisA. The regression equation for the simple version of ProxyVis that uses a *single regression* for clouds at all levels, is given by

$$PV_{SA} = X_{SA} C07^5, \quad (9)$$

where X_{SA} is the normalized coefficient (Table 6), and the predictor mean and standard deviation are listed in Table 5.

2) PROXYVIS—POSTPROCESSING

A few postprocessing steps were added to further improve the appearance of the product. Postprocessing is applied to each of PV, PV_A , PV_S , and PV_{SA} to obtain the four versions of ProxyVis: PrVis, PrVisA, PrVisSimple, and PrVisSimpleA, correspondingly. Here we provide the derivation to obtain the main PrVis product from PV defined in (4).

To estimate PrVis, the minimum value of 3.0×10^{-5} is used for the term $|C11 - C07|$ to avoid large negative values of its natural log. For the *GOES-16* full disk, approximately 1%–2% of the pixels can still yield negative PV in (4). Thus, we define

$$PV_{\text{positive}} = \begin{cases} 0, & \text{where } PV < 0 \\ PV, & \text{where } PV \geq 0 \end{cases}, \quad (10)$$

and estimate PrVis as normalized PV_{positive} to ensure that all PrVis values are between 0 and 1:

$$PrVis = \left[\frac{PV_{\text{positive}} - \min(PV_{\text{positive}})}{\max(PV_{\text{positive}}) - \min(PV_{\text{positive}})} \right]^{2/3}. \quad (11)$$

Taking the power of (2/3) in (14) helps to make the final GeoProxyVis product more continuous across the day/night terminator when day- and nighttime parts are combined (see section 2i). For *GOES-16/18* $\min(PV_{\text{positive}}) = 0$ and $\max(PV_{\text{positive}}) = 0.7$ (0.84 for *GOES-18*), are representative for the full disk.

f. Comparison of the four versions of ProxyVis

PV_L , PV_H , and PV from (2), (3), and (5) explained 45.7%, 39.6%, and 68.7% of the variance of low, high, and all clouds from the developmental dataset, respectively (Table 6). For the single-regression PV_A for all clouds the $R^2 = 68.2\%$,

which is slightly lower than for two-regression version. Based on those numbers and visual analysis of different versions for multiple ProxyVis scenes (not shown), it was decided to use PrVis (11), the multichannel two-regression version, as the main algorithm that is demonstrated here and used for all ProxyVis figures unless stated otherwise. The threshold BT_{tr} (1) between PV_L and PV_H (2), (3) was empirically selected to optimize the performance for low clouds, at the expense of the performance for high clouds, consistent with the goal of ProxyVis to emphasize low-level clouds.

The normalized regression coefficients from Table 6 show that the VIIRS 3.74- μm channel I04 provides the largest contribution to PrVis. This enables the development of single-channel simple versions of ProxyVis (PrVisSimple and PrVisSimpleA) using VIIRS channel I04 or a similar 3.9- μm channel common on geostationary sensors. Similar to the main algorithm, the simple version that uses separate equations for low and high clouds performs slightly better. The corresponding R^2 values for PrVisSimpleA are 65.7% for a single regression for all clouds, and for the two-regression version PrVisSimple, 66.5% for all, 41.7% for low, and 35.2% for high clouds. These values are slightly lower than those of the main ProxyVis algorithms. A visual comparison of different versions of ProxyVis (e.g., Fig. S1 in the online supplemental material) confirmed that despite different statistics (Table 6) for many scenes all four versions of ProxyVis, including the simple single-channel versions (PrVisSimple and PrVisSimpleA) and the main versions (PrVis and PrVisA), look very similar.

Each ProxyVis algorithm version has a specific utility, and users can choose the version of ProxyVis that best matches their application (Table 4). The single regression equation versions are preferable if continuity is essential (e.g., if used as input to other applications, such as ML or other multispectral products) and might work best for observing midlevel clouds. The two-regression versions are preferable if accuracy of the low cloud detection is a priority. The simple single channel versions (PrVisSimple and PrVisSimpleA) are needed so that ProxyVis can be applied to older GOES satellites, to *GOES-17* during the time when it is affected by the loop heat pipe issues (STAR 2019; Goodman et al. 2019; Van Naarden and Lindsey 2019), and for cases before the current generation of geostationary satellites. For example, PrVisSimple is being considered for operational implementation for *GOES-13* and *GOES-15* that are still used by JTWC. Based on NWS forecaster feedback, the simple versions also might work better for MSG SEVIRI. Further, the simple single-channel versions can be used to develop long ProxyVis datasets that combine

current and previous generation GOES data and may be needed for some ML applications.

g. Evaluation of the functional form of the prediction equations

The channels used in ProxyVis are commonly used for cloud detection (Schmit et al. 2005). The shortwave IR $3.74\text{-}\mu\text{m}$ ($3.9\text{-}\mu\text{m}$) channel is more sensitive to warm temperatures than the longwave IR (owing to the Planck function), and is also less sensitive to intervening cirrus clouds, which are more forward-scattering at $3.9\text{-}\mu\text{m}$ compared to the longwave IR where cirrus clouds are more absorbing (Baran et al. 1999; Schmit et al. 2005; Velden et al. 2005). The combined effects make the $3.9\text{-}\mu\text{m}$ channel one of the most useful channels for identifying near-surface features such as low-level clouds and underlying SSTs.

The SWD, M15 ($10.76\text{ }\mu\text{m}$) – M16 ($12.01\text{ }\mu\text{m}$), provides information about the lower-tropospheric water vapor distribution (Prabhakara et al. 1974; McMillin 1975; Lindsey et al. 2014). This sensitivity allows the differentiation between dry and moist environments in cloud-free areas. The SWD is positive for clear-sky moist areas and thin cirrus, close to zero for dry areas and thick clouds, and negative for volcanic ash and most blowing dust. The absolute value of SWD used in PrVis cannot differentiate between dust and thin cirrus clouds; however, it can still differentiate between thick clouds (deep convection) and thin cirrus.

Finally, the difference [$\text{IO4 } (3.74\text{ }\mu\text{m}) - \text{M14 } (8.55\text{ }\mu\text{m})$] is related to the difference used in the fog product, [$10.7\text{ } (11.2) - 3.9\text{ }\mu\text{m}$; Ellrod 1995] and highlights the emissivity differences for land versus ocean surfaces as well as ice versus liquid water. The fog product is positive for liquid clouds, close to zero for ice clouds and clear areas, and negative for thin cirrus. The absolute value used in PrVis cannot differentiate between thick clouds and cirrus; however, it can still distinguish liquid water and ice clouds. Only 7.6% or less of the SWD and fog product equivalent used in PrVis (Table S1 in the online supplemental material) are negative, thus most of the information is preserved in PrVis.

To assess the independence of the predictors, Table 7 shows the correlation matrix for the terms used in PrVis (11). While all correlations are significant at the 99.9th percentile, no variable explains more than about half of the variance of another variable. The largest correlation, 0.75, is observed between the terms $|\text{C13} - \text{C15}|^{0.4}$ and $\ln(|\text{C11} - \text{C07}|)$ for the low clouds. The very low p values for the regression also confirm that all terms are highly significant (at 0.01% or less), with most of the contribution coming from C07^5 , followed by $|\text{C13} - \text{C15}|^{0.4}$, and with the least contribution from the $\ln(|\text{C11} - \text{C07}|)$ term. Figure 3 shows the BT of VIIRS channel $3.74\text{ }\mu\text{m}$ (Fig. 3a), the 5th power of that BT (Fig. 3b), and the final PrVis (Fig. 3c) versus scaled DNB for scene 1 in Table 3. The 5th power of $3.9\text{ }\mu\text{m}$ is more linearly related to DNB compared to $3.9\text{ }\mu\text{m}$, and the relationship between PrVis and DNB is even more linear. Thus, using nonlinear transformation of independent variables allows us to linearize the relationships between our independent variables and the scaled DNB, which improves ProxyVis similarity to DNB.

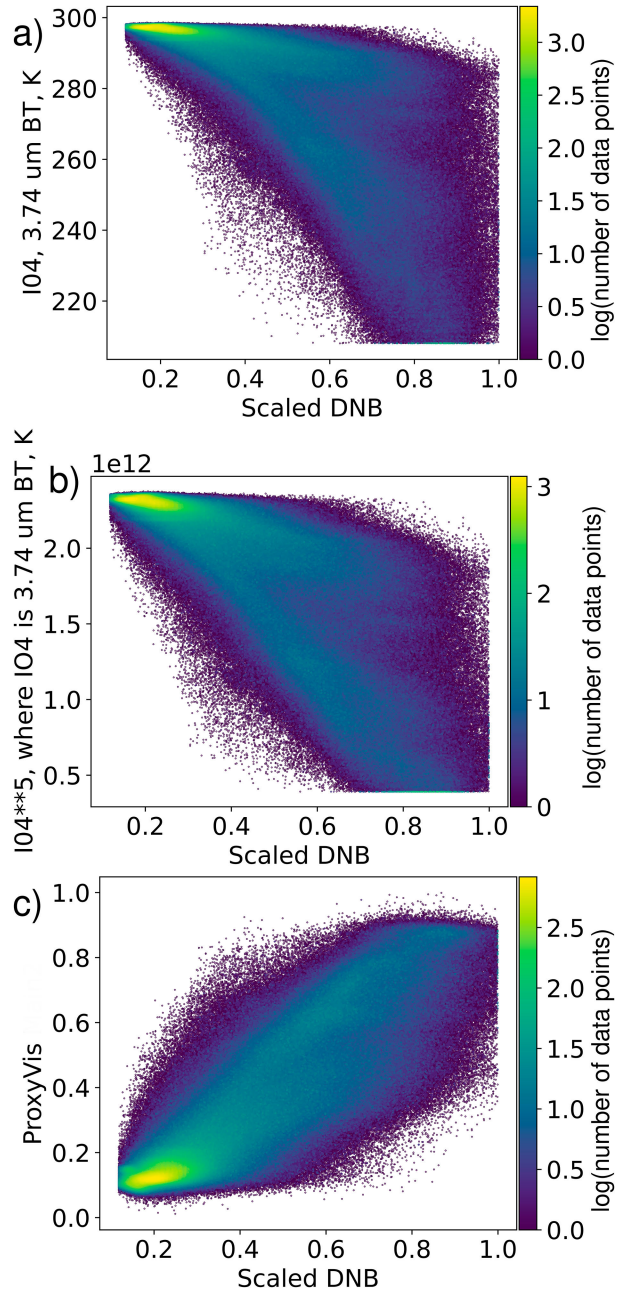


FIG. 3. Scatterplots of (a) VIIRS BT for channel IO4, $3.74\text{ }\mu\text{m}$, (b) VIIRS $(\text{IO4})^5$, and (c) PrVis vs scaled DNB, for Tropical Storm Ida on 26 Sep 2015 (scene 1 from Table 3). The plots illustrate how taking the fifth power of BT of channel IO4, and further estimating ProxyVis (PrVis) makes the relationship between DNB and the resulting function more linear. The color shading shows the density of the points. To estimate density, all points were split into 1000 bins of the 2D histogram, and the density of points in each bin was evaluated. The points were then sorted to make sure the densest points are plotted last. The color indicates the common logarithm of the number of data points used to make each point on the scatterplot.

TABLE 8. As in Table 3, but for VIIRS scenes used for independent verification of ProxyVis imagery.

Scene No.	Storm ID/name, latest best track intensity (kt)	Area bounds	Start time (UTC time + date)	End time (UTC time and date)	Percent full and closest full moon
1	al092019/Humberto, 25 kt	0°, 40°N, 90°W, 50°W	0637:338 UTC 14 Sep 2019	06:44:396 UTC 14 Sep 2019	Full, 14 Sep
2	ep132019/Kiko, 105 kt	10°N, 40°N, 140°W, 105°W	0936:223 UTC 15 Sep 2019	09:50:352 UTC 16 Sep 2019	98%, 14 Sep
3	wp112019/Krosa, 45 kt	5°N, 40°N, 127°E, 160°E	1618:285 UTC 14 Aug 2019	16:32:413 UTC 14 Aug 2018	99%, 15 Aug
4	io022019/Vayu, 35 kt	5°S, 24°N, 40°E, 70°E	2131:122 UTC 16 Jun 2019	21:45:232 UTC 16 Jun 2019	99%, 17 Jun

h. Independent verification

Independent verification of PrVis was conducted using VIIRS data for four scenes from four different basins (Table 8), selected using the same criteria as for the training scenes (section 2c). Table 9 shows that for all four independent scenes, the R^2 is higher than for the dependent sample for both low and high clouds, with the values ranging from 50.7% to 70.8% for low and from 50.3% to 66.8% for high clouds (corresponding dependent values are 45.7% and 39.6% for the low and high clouds, respectively). The comparison of VIIRS DNB with PrVis for independent scenes 2–4 is shown in Fig. 4. (The comparison for independent scene 1 is shown later in Fig. 14.) The total R^2 is higher than the dependent sample for the North Atlantic, western Pacific, and Indian Ocean (77.6%–85.2% versus 68.7% for the dependent sample) and is slightly lower (66.8%) for the eastern Pacific. The comparable or higher R^2 values for the independent data confirm the strengths of the relationships from the training scenes. Lower R^2 were found in the eastern Pacific and at higher latitudes in the Atlantic, which helped to identify a limitation of the current algorithm for areas with cold SSTs (see section 4).

i. Steps to combine nighttime ProxyVis with daytime visible imagery

To best display the full disk imagery, the ProxyVis generated from 2-km resolution ABI channels is regridded to the native 0.5-km grid of ABI C02, 0.64 μm . The regridding is done using nearest-neighbor interpolation to avoid generating nonexistent clouds that could be produced when multiple points are used to generate the interpolated value. Further, the visible imagery is adjusted to provide a seamless day/night transition. Following Minnis and Harrison (1984) and Tsonis and Isaac (1985), a gamma correction is applied to visible

channel C02, and to brighten visible imagery in low sun angle conditions the entire visible image is normalized using

$$\text{VisDisp} = \left(\frac{\text{Vis}}{\cos\theta} \right)^\gamma, \quad (12)$$

where θ is the solar zenith angle, $\gamma = 0.5$, and Vis is the reflectance for the ABI C02. The final operational GeoProxyVis imagery combines nighttime PrVis and daytime VisDisp into a single product. Both day and night parts are adjusted to be between 0 and 1.3 to match the range of visible C02 in the operational AWIPS2 (2019):

$$\text{VisDisp}(\text{VisDisp} > 1.3) = 1.3, \quad (13)$$

$$\text{GeoProxyVis} = \begin{cases} 1.3 \text{ PrVis,} & \text{if } \theta > 89^\circ \\ \text{VisDisp,} & \text{if } \theta \leq 89^\circ \end{cases}. \quad (14)$$

As shown in Fig. 5 for a GOES-16 scene primarily containing low-level clouds, the transition between the daytime and nighttime parts of the GeoProxyVis imagery is very smooth, which allows for the tracking of cloud features across the terminator.

3. Results

The results described in this section include: 1) comparison of PrVis with legacy products (e.g., stretched 3.9- μm and fog), and 2) independent examples of GOES-16 ProxyVis imagery.

a. Comparison with legacy products and DNB with low moonlight

Figure 6 shows a comparison of DNB (Fig. 6a), and nighttime visible-like products estimated from VIIRS IR data, including PrVis (Fig. 6b) and legacy products used for tracking low-level clouds, the fog product (Fig. 6c) and the stretched

TABLE 9. Variance explained for the independent VIIRS scenes used for ProxyVis verification.

Scene No.	Storm ID/name, latest best track intensity (kt)	R^2 single regression PrVisA	R^2 two regressions PrVis	R^2 low clouds for PrVis	R^2 high clouds for PrVis
1	al092019/Humberto, 25 kt	74.2%	77.6%	70.8%	58.5%
2	ep132019/Kiko, 105 kt	66.0%	66.8%	50.7%	56.1%
3	wp112019/Krosa, 45 kt	80.5%	79.8%	55.3%	50.3%
4	io022019/Vayu, 35 kt	84.6%	85.2%	62.2%	66.8%

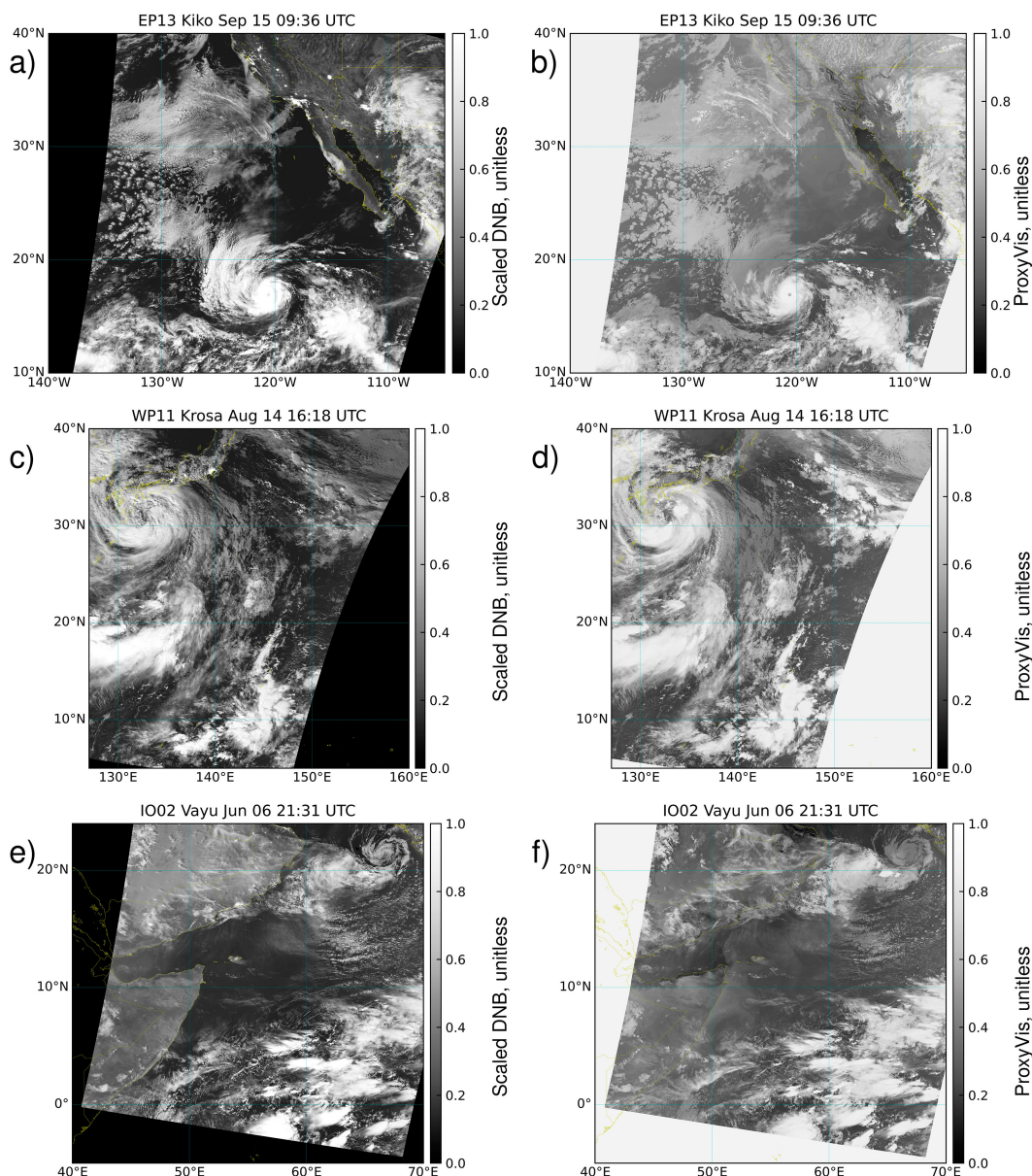


FIG. 4. Comparison of VIIRS DNB scaled using (a),(c),(e) gamma-DNB scaling with (b),(d),(f) PrVis for independent scenes 2 (Hurricane Kiko), 3 (Typhoon Krosa), and 4 (Tropical Cyclone Vayu) from Table 8. Note that the plots show all available data; however, for verification purposes, we removed land points using the same procedure that was used to remove land points from the training dataset.

$3.74 \mu\text{m}$ (Fig. 6d). PrVis does not exactly replicate the DNB image, but it improves the contrast of the low-level clouds compared to legacy products and shows most of the low-level clouds seen in the DNB image. That detail is highlighted in the area inside the yellow rectangle, where the stretched $3.74 \mu\text{m}$ saturates, and the fog product struggles due to water vapor masking of low clouds. These differences can be seen more clearly in Fig. 7, which zooms into that area.

As described earlier, one of the DNB limitations is that its quality is a strong function of the moon phase. While near the full moon (Figs. 6a,b, the moon is 96% full) DNB and PrVis

look similar, near the new moon (Fig. 8, the moon is 33% full) the quality of DNB degrades significantly, while ProxyVis is not affected.

b. GOES-16 examples

The main advantage of ProxyVis is that it can be applied to geostationary satellite data from all current and many past satellites to create animations. This subsection presents examples of GeoProxyVis imagery that is used by NWS and JTWC. For these examples, we use *GOES-16* data.

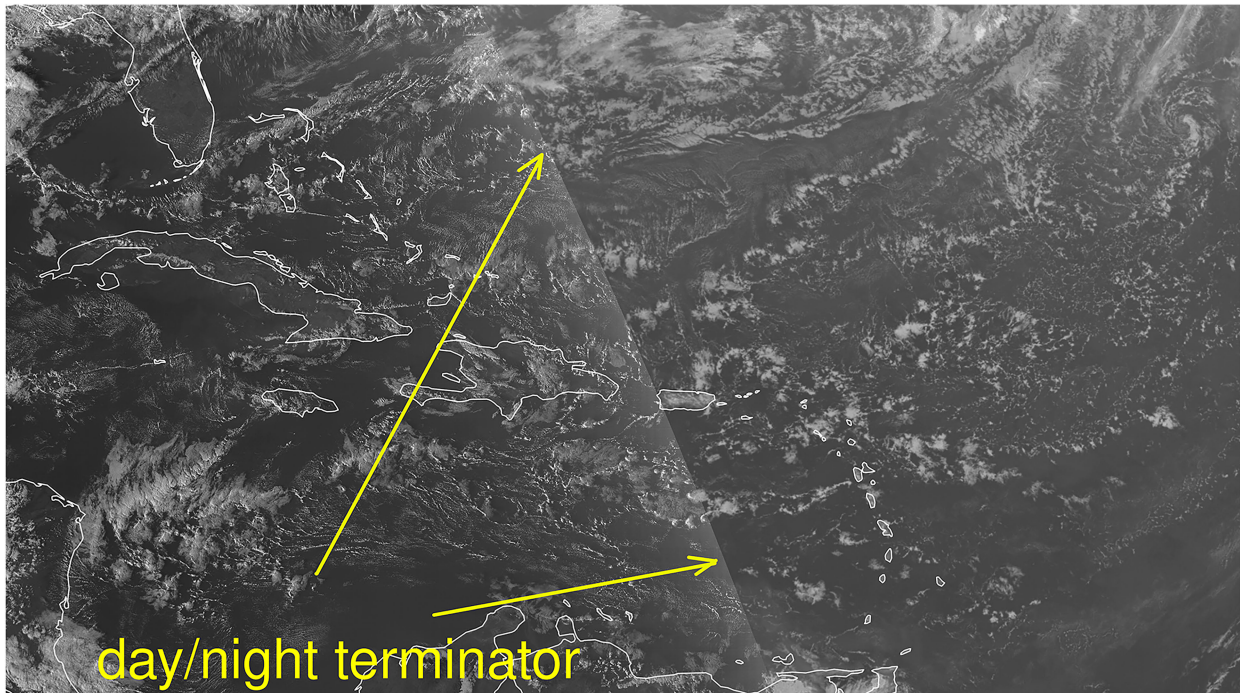


FIG. 5. Day/night terminator (shown by yellow arrows) as seen on *GOES-16* GeoProxyVis at 2210 UTC 29 Jan 2023. The image is centered near 20°N, 65°W. Visible *GOES-16* C02 (VisDisp) is on the left, and PrVis is on the right.

Figure 9 shows a GeoProxyVis [Hovmöller \(1949\)](#) diagram demonstrating the advantages of ProxyVis for manual feature tracking. Cloud features highlighted in Fig. 9 include a line of stratocumulus clouds in the eastern North Atlantic moving equatorward in the northeast trade winds, and the cloud lines that provide the direction of the low-level flow in the eastern Pacific. The smaller-scale features are associated with a larger-scale, westward-moving tropical wave. Note that the day/night terminator highlighted in red is not evident.

Another GeoProxyVis example (Fig. 10) shows subregions from the nighttime portion of the full disk image at different zoom levels. The full disk GeoProxyVis (Fig. 10a) shows features that have a similar appearance to visible imagery. These include synoptic-scale midlatitude weather systems (cutoff lows in the Southern Hemisphere, a frontal system stretching across the eastern United States), stratocumulus decks off the South American coast, and a prominent ITCZ in the eastern Pacific. The zoom on the Atlantic (Fig. 10b) again looks similar to visible and shows easterly waves (east of the Lesser Antilles), cold fronts (Texas coast), cloud lines, boundary layer rolls, and mesoscale convergence zones. Finally, in the Gulf of Mexico subsector (Fig. 10c) one can more clearly see the boundary layer rolls (central Gulf of Mexico), cloud lines (e.g., north coast of Cuba), and the ragged appearance of the cold frontal boundary. At this higher resolution, additional features can be seen such as the cold SST wake from Hurricane Michael between Cuba and Appalachicola, Florida, and the cold air behind the cold front. The cold ground signature slightly reduces the contrast with low-level cloud features.

An accurate estimate of the TC center location is critical for determining the initial motion of the system, which serves as an input to the guidance models used for forecasting TC track, and for the accurate estimation of TC intensity via the Dvorak technique, particularly when the shear and embedded center patterns are used ([Dvorak 1984](#)). Errors in both the initial position and intensity estimates of the TC can result in poor track and intensity forecasts, especially in the short term. With ProxyVis, the LLCC can be identified almost as well as in DNB, even in the presence of cirrus clouds. Figure 11a shows an example when the LLCC cannot be easily determined from the longwave IR image for Hurricane Otis as it was being sheared from the northeast, but is clearly seen in PrVis (Fig. 11c) and DNB (Fig. 11b). Additional examples of animated GeoProxyVis loops are provided in the supplemental material and are also available in real-time on SLIDER ([SLIDER 2023](#); [Micke 2018](#)).

4. Use in NWS and JTWC operations

This section discusses 1) operational implementation of ProxyVis imagery at NWS and JTWC; 2) operational use at NHC; 3) operational use at JTWC, and 4) ProxyVis strengths and limitations based on forecaster feedback.

a. Overview of the operational implementation

Due to the simplicity of the formulation, ProxyVis is easy to implement in different programming languages and platforms. That facilitates the transition to NWS operations, which often have significant software constraints. *GOES-16* GeoProxyVis

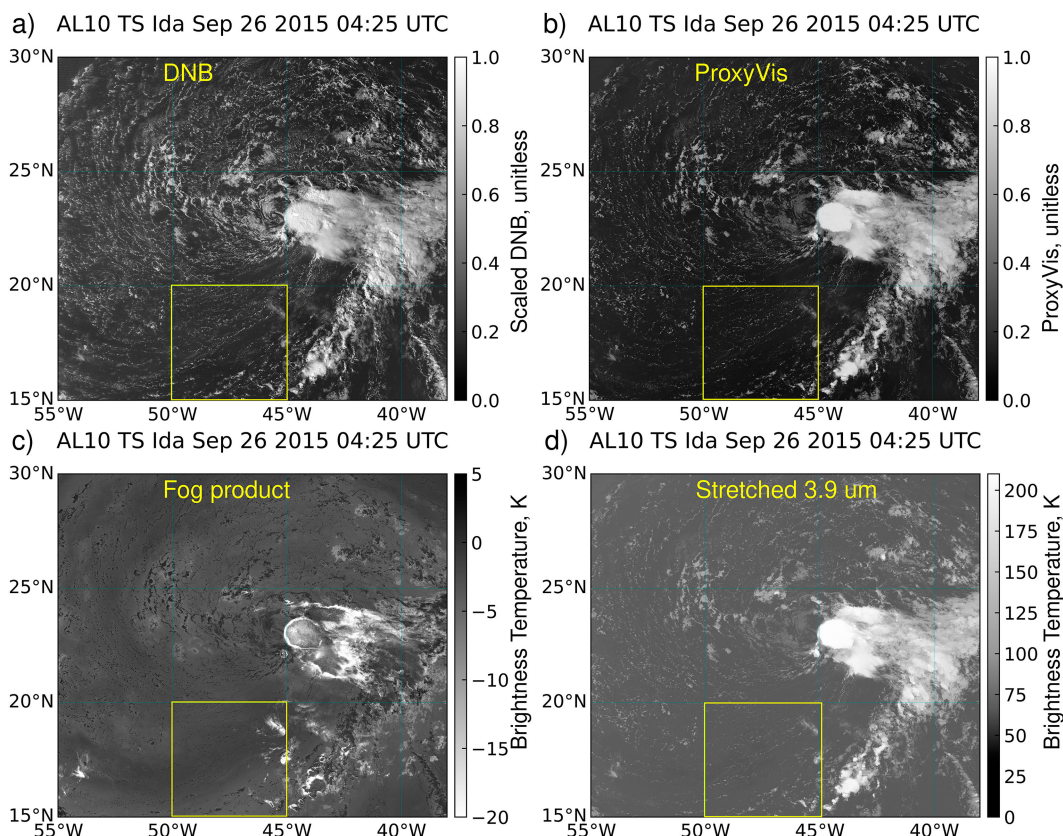


FIG. 6. VIIRS data at 0425 UTC Sep 2015 (scene 1 in Table 3): (a) scaled DNB, (b) PrVis, (c) fog product ($10.76 - 3.74 \mu\text{m}$) with low-level clouds corresponding to more positive values shown in black, and (d) $3.74\text{-}\mu\text{m}$ stretched (VIIRS channel similar to $3.9 \mu\text{m}$) to enhance low-level clouds, for the same scene as in Fig. 3. The yellow square highlights low-level clouds that are seen in (a) and (b), but are hard to see in (c) and (d). In (c), the BT difference from -20 to $+5$ K that captures the differences between high and low clouds is shown. The full range of the image in (c) is from -59.8 to 41.4 K, with 0.46% of BT differences less than -20 K and 0.06% of BT differences larger than 5 K.

imagery first became available at NHC in real-time in NAWIPS in the spring of 2018, where the product was generated at CIRA and sent to NHC via a Local Data Manager feed. Starting with the 2019 hurricane season, GeoProxyVis became part of the standard NHC operational satellite data suite that is used routinely by both TAFB and HSU as one of the four main operational satellite products and replaced the legacy fog product. For the 2020 Atlantic hurricane season, GeoProxyVis was implemented in NHC's in-house AWIPS2 (2019) via ISATSS (Guillot et al. 2017) for GOES-16/17 full disk. In 2021, 5-min CONUS and 1-min mesoscale sectors were added to ISATSS implementation. NHC now routinely archives the GeoProxyVis data as a replacement for visible imagery and uses it for post-storm analysis and writing their annual TC reports.

OPC and WPC have been using GeoProxyVis provided by CIRA in NAWIPS since 2020. The WPC International Desk has been using GeoProxyVis on SLIDER since 2020. In 2022, GeoProxyVis was implemented in the ATCF via the NRL GeoIPS (GeoIPS 2023; Surratt et al. 2016), and the animated version was demonstrated to JTWC forecasters via SLIDER and AWIPS2. NWS is working on making ProxyVis available to NWS WFOs and other National Centers, including Pacific

Region, Alaska Region, and CPHC. The Australian Bureau of Meteorology has also expressed interest in using ProxyVis in its operations. The real-time 10-min (15-min for MSG) full disk animated GeoProxyVis for GOES-16/17(18) has been available on SLIDER since the end of 2017 (2022), for Himawari-8(9) since 2021 (2022), and for Meteosat-9/10/11 since 2022. The operational use of GeoProxyVis was mentioned in the IWTC-10 report (Herndon et al. 2022). Given the differences between geostationary satellites for which ProxyVis is currently implemented, we are confident that ProxyVis can be implemented for Geo-KOMPSAT-2A and MTG, as well as for next-generation geostationary satellites such as GeoXO (NOAA/NESDIS 2022).

b. Use in NHC operations

For HSU forecasters, a critical part of the forecast process is the analysis of the TC center location and its intensity and structure at 0000, 0600, 1200, and 1800 UTC. ProxyVis imagery is one of the most useful imagery products for estimating the TC center location, and offers significant improvements over longwave IR as it allows forecasters to identify low-level cloud lines. Animation of ProxyVis images provides key

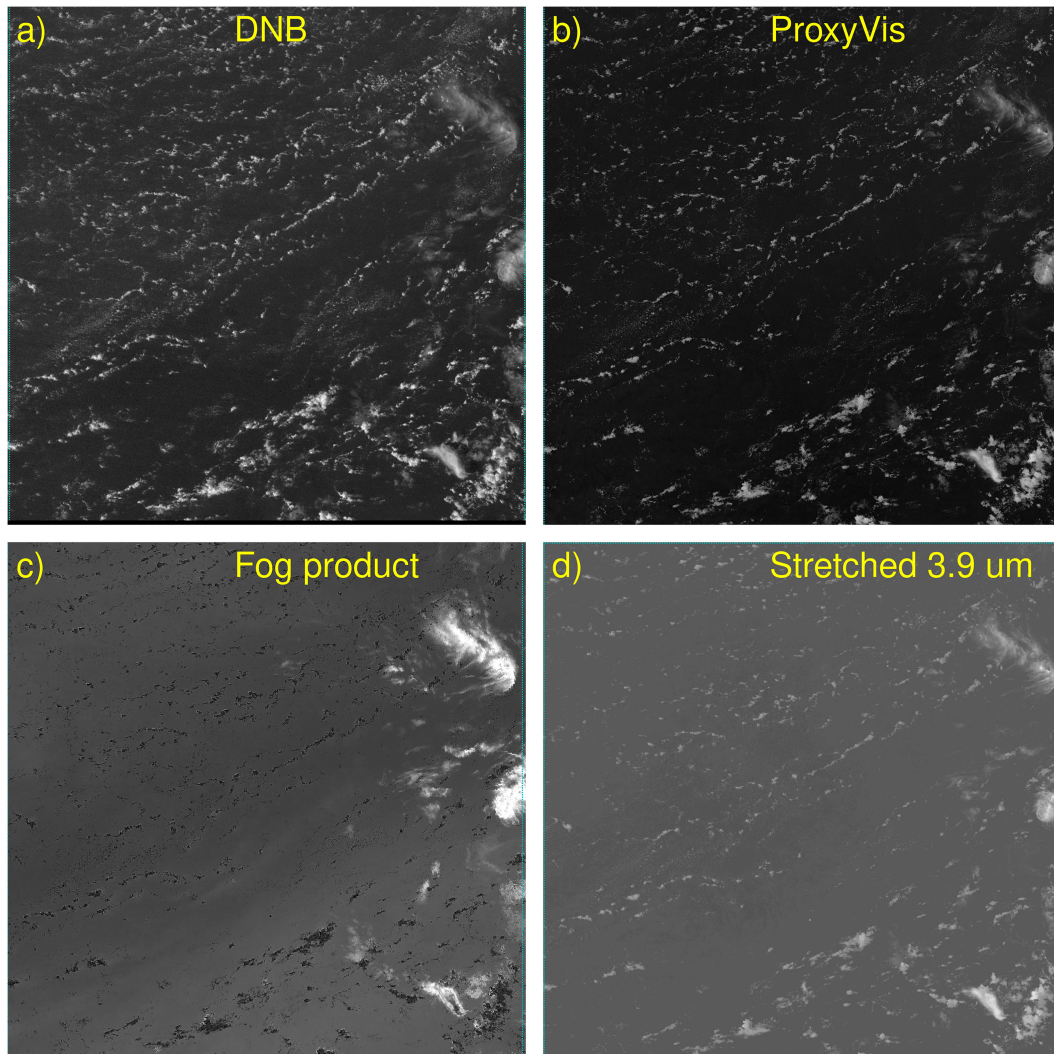


FIG. 7. As in Fig. 6, but only the part highlighted by the yellow square in Fig. 6 of each panel is zoomed in on.

information for determining the location of the TC's surface circulation, which is especially helpful for weaker TCs that do not have an eye or an otherwise easily identifiable center in longwave IR imagery. ProxyVis imagery is also very useful in the genesis stage for helping to determine if a disturbance has a closed circulation. NHC forecasters also found that ProxyVis imagery significantly increased situational awareness during the overnight hours; allows tracking the development of small clusters of convection that may be warm topped and not evident on other IR channels; helps to identify convection associated with tropical waves; is useful for determining the location of the ITCZ or weak fronts that do not have deep convection; and is helpful for postseason TC analysis.

Figure 12a shows GeoProxyVis imagery just before 1000 UTC 7 July 2018 over Tropical Depression Three (which later became Hurricane Chris). For this poorly organized system with a broad circulation, ProxyVis was very useful to the NHC forecaster in locating/tracking the center and establishing the initial motion estimate, as noted in the NHC's TC Discussion issued at 5:00 a.m.

EDT 7 July 2018 (NOAA/NHC 2018b). ProxyVis better emphasizes low clouds and looks more like visible imagery compared to the fog product (Fig. 12b) that was operational at NHC before ProxyVis became available. Another example of real-time use of ProxyVis imagery is found in the NHC's TC Discussion for Tropical Storm Florence (Fig. 13) issued at 0500 AST 7 September 2018, where the NHC forecaster stated: "Nighttime Proxy-Vis and earlier microwave imagery indicate that Florence has turned westward, with an estimated initial motion of $275^{\circ}/6$ kt" (1 kt ≈ 0.51 m s $^{-1}$) (NOAA/NHC 2018a).

c. Use in JTWC operations

GeoProxyVis was implemented at JTWC via both AWIPS2 and ATCF (see section 4a) for the latter half of the 2022 TC season. Initially considered an experimental tool to assist in the nighttime positioning of LLCC, ProxyVis quickly became a favored tool in the satellite analyst's toolbox for cloud feature identification across the Pacific and Indian Oceans. The

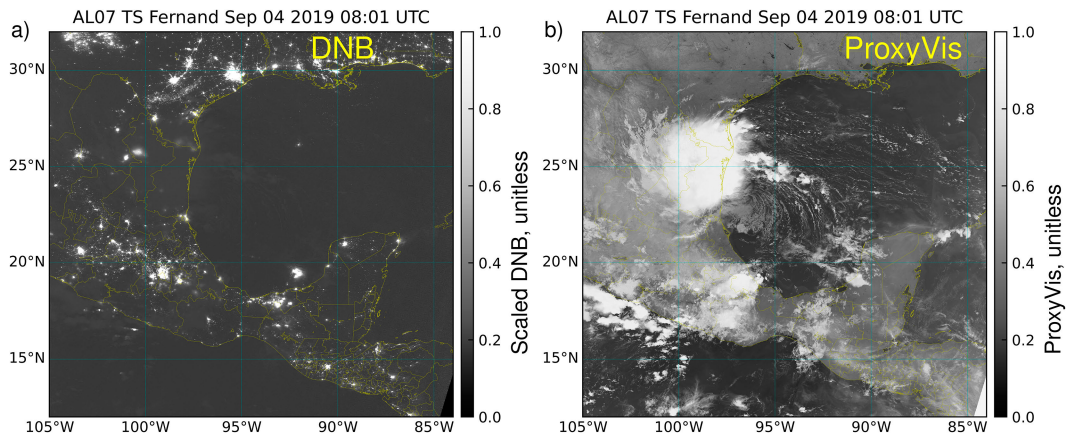


FIG. 8. Comparison of (a) scaled VIIRS DNB and (b) VIIRS PrVis for Atlantic Tropical Storm Fernand (2019) at 0801 UTC 4 Sep 2019. It is hard to see the clouds in the VIIRS DNB image because the moon is only 33% full, and, in addition, there is a large number of city lights in the scene. ProxyVis works the same way independently on the moon phase and the presence of city lights and other non-cloud-related light sources. The exact appearance of the DNB image is a function of the scaling chosen for the DNB; however, the visibility of clouds is always degraded closer to the new moon.

capability to view looping full disk imagery on AWIPS2, while easily zooming for feature analysis, enhanced JTWC's ability to interrogate TCs and genesis areas for development.

Feedback from JTWC analysts and forecasters includes accolades such as "When Vis is not available, this is my Go-to." ProxyVis was found to be a valuable tool to assist when compared to the same analysis performed using previously approved procedures, and the GeoProxyVis loop is now utilized day and night.

d. ProxyVis strengths and limitations

Considerable feedback was obtained from NHC, OPC, WPC, and JTWC forecasters during the GOES-R PG (Goodman et al. 2012) evaluation in 2019–22 through surveys and informal communications. Forecaster feedback on ProxyVis was generally very positive and indicated that ProxyVis is a vast improvement over the legacy products, and provides much better temporal and spatial coverage compared to OLS and DNB. According to coauthor John Beven, a senior NHC/HSU forecaster, ProxyVis imagery is "... *perhaps the most useful multi-spectral product ... which provides almost visible-quality imagery at night.*" NHC TAFB forecasters indicated that ProxyVis is useful for multiple meteorological situations, including determining LLCC for Dvorak estimates and identifying the location of fronts and tropical waves and areas with fog and low stratus. Ideas for improvements (summarized below) were usually provided informally. Direct quotes from NHC and JTWC forecasters are included in the online supplemental material.

The original purpose of ProxyVis was to replicate visible imagery at nighttime with the main emphasis on capturing nighttime oceanic low-level clouds that cannot be seen in conventional satellite IR imagery. After ProxyVis became routinely available, forecasters started using it in a variety of conditions and synoptic situations for which the product was not originally designed. For example, forecasters found ProxyVis

useful for consistently tracking low-level circulations from daylight to darkness, and for tracking high-latitude cloud features as far north as the Bering Sea. Despite the degraded quality, forecasters also found ProxyVis useful over land.

The primary limitations noticed by forecasters follow from the ProxyVis algorithm design. First, all four ProxyVis algorithms (PrVis, PrVisA, PrVisSimple, and PrVisSimpleA) are only valid during the nighttime and have not been evaluated or verified during the daytime, for $\theta \leq 89^\circ$. Most of the ProxyVis signal is coming from the $3.9\text{-}\mu\text{m}$ channel (section 2g) that has both reflective and emissive components, thus ProxyVis will behave differently during daytime. GeoProxyVis slightly degrades for large satellite zenith angles; however, most of that degradation comes from the visible part (e.g., Zhuge et al. 2012) of GeoProxyVis imagery. ProxyVis was primarily developed for use by the TC forecast centers, and the training data only included over-the-ocean data and did not include any high-latitude scenes. Thus, ProxyVis performance over land and at higher latitudes is expected to be degraded. Sometimes (not shown), clouds seen over land in the longwave IR imagery cannot be seen in ProxyVis. Further, even though statistical techniques were used for development, ProxyVis imagery was designed as a qualitative product. ProxyVis was trained on dynamically scaled DNB, and the final ProxyVis formulation was selected based on subjective evaluation by the developers, similar to how other multispectral products are created [e.g., RGBs or GeoColor (Miller et al. 2020)]. Using DNB reflectance (Miller and Turner 2009) could be a cleaner way to train the algorithm. The use of two regressions in PrVis improves the display of low clouds at the expense of the degraded presentation of high- and midlevel clouds and sometimes creates a visible boundary between high and low-level clouds. Also, sometimes VisDisp saturates in the transition region between ProxyVis and visible imagery. The limitations described above will be addressed in future work.

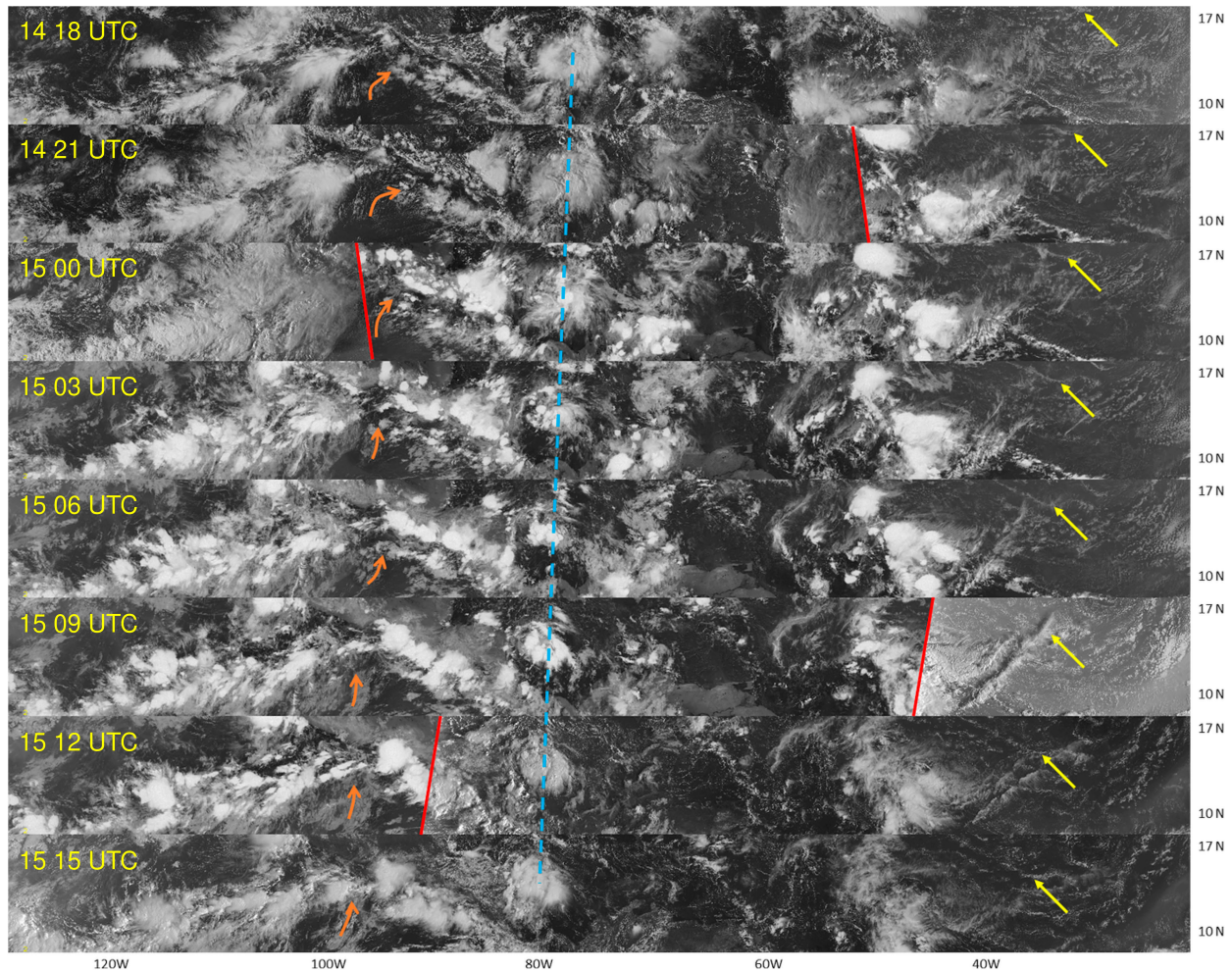


FIG. 9. Longitude vs time cross section of the *GOES-16* GeoProxyVis. The images are 3-hourly (top) starting at 1800 UTC 14 Oct and (bottom) ending at 1500 UTC 15 Oct 2018, centered at 13°N. The images have been remapped to a Mercator projection before being added to the individual time panels. The day/night terminator is indicated by the solid red line. Multiple synoptic features can be seen in these images, including some stratocumulus clouds moving equatorward in the northeast trade winds in the eastern part of the scene (yellow arrows), a convective system moving to the west (dashed blue phase line), and a reversal of low-level winds from westerly to southerly indicated by low-level cloud lines in the eastern Pacific (orange arrows).

The verification on independent VIIRS imagery (section 2h) showed that for independent scenes the R^2 was comparable to or greater than that for the dependent data. However, when the verification sample was stratified by latitude, R^2 decreased sharply in the North Atlantic, north of $\sim 40^\circ\text{N}$ where the SST is quickly decreasing. To investigate that issue further, the Atlantic scene (Table 9) was extended to 50°N , and the R^2 was calculated for latitudes from 40° to 50°N . Results showed that the R^2 was much smaller for that region, with values of 4.6% for low clouds, 17.9% for high clouds, and 58.2% for all data. As shown in Fig. 14, in this area there are large cloud-free areas over cold SST that look black in the DNB image but show the SST in ProxyVis, which significantly reduces the correlation. Although not as dramatic as for the North Atlantic region, the R^2 values for the eastern Pacific scene in Table 9 are also reduced, likely

due to much colder SST north of 25°N , and the large amount of marine stratocumulus clouds that were not included in the training scenes.

The SST signal in ProxyVis is a by-product of using the IR channels and is reduced compared to legacy products. This SST signal slightly affects the low-level cloud visibility over cold SSTs (north of 40°N in the Atlantic), but does not affect it at low latitudes (in the Gulf of Mexico or in the eastern Pacific, Fig. 15a). Forecasters noted that SST features can help with increased situational awareness, and in the animated geostationary loops are easy to distinguish from clouds based on the time evolution. The SST signal, however, could be an issue for using ProxyVis as a nighttime replacement for visible channels in multichannel products such as RGBs or GeoColor (Miller et al. 2020). Due to the use of the fifth power of the $3.9\text{-}\mu\text{m}$ channel, additional ProxyVis by-products

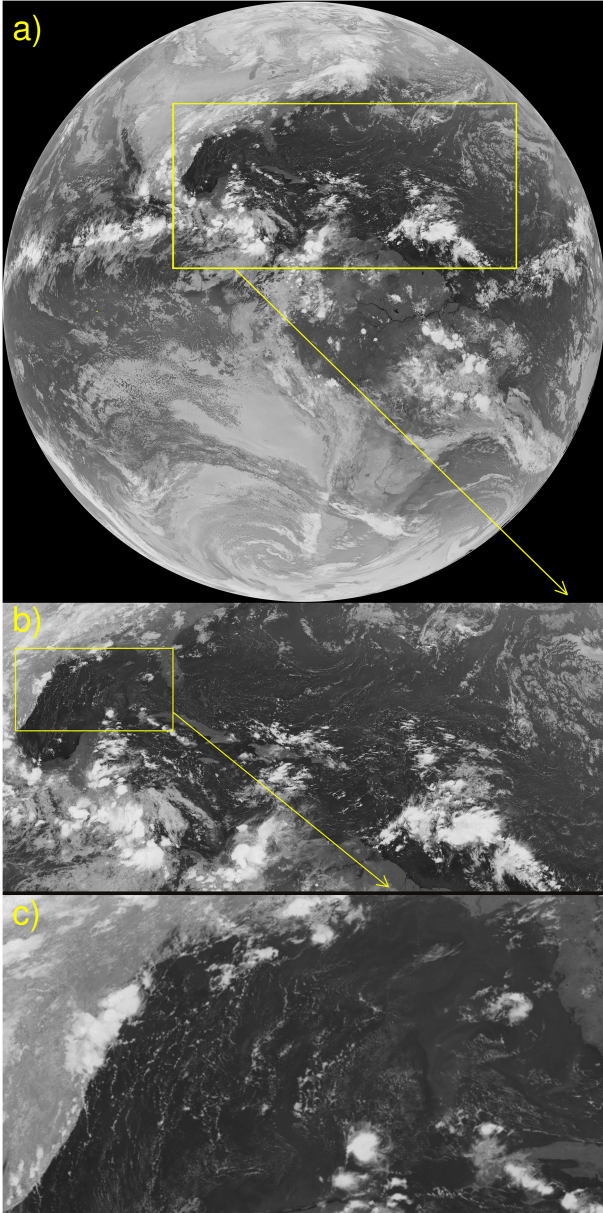


FIG. 10. *GOES-16* GeoProxyVis for (a) full disk, (b) an Atlantic subsector, and (c) a Gulf of Mexico subsector at 0515 UTC 16 Oct 2018 when the entire full disk is under darkness.

are the nighttime wildfires, where ProxyVis shows both nighttime wildfires and clouds, and heat islands (not shown). ProxyVis also captures ship tracks shown in Fig. 15b.

5. Summary and future work

The nighttime ProxyVis imagery is enabled by the existence of the JPSS VIIRS DNB imagery and is an improved satellite product for tracking low-level clouds at night. While other multichannel products for highlighting low-level clouds, such as the nighttime microphysics RGB, might be sometimes

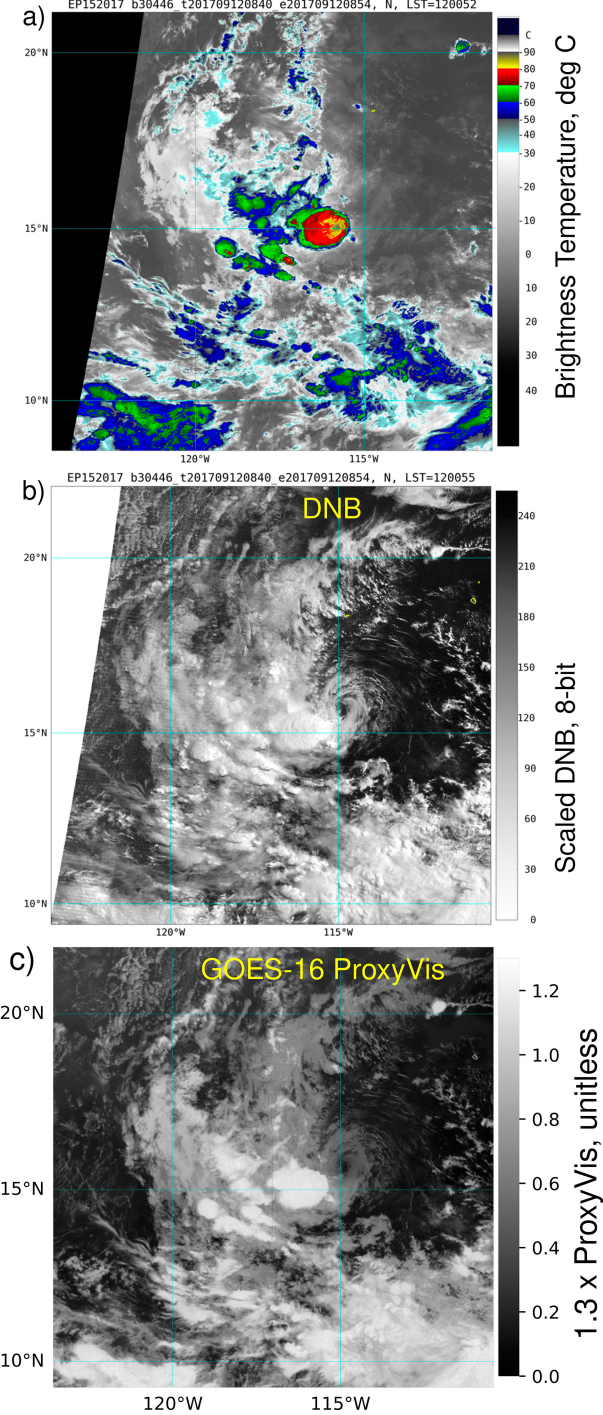


FIG. 11. (a) Color-enhanced VIIRS IR image (105, 11.45 μm) and (b) gamma-DNB-scaled DNB image of the disturbance that became Hurricane Otis at 0845 UTC 12 Sep 2017. A standard IR enhancement does not show the warmer low-level clouds around the exposed LLCC depicted in the DNB image. (c) *GOES-16* PrVis at 0900 UTC 12 Sep 2017. The LLCC could be seen in ProxyVis almost as well as in the DNB image. Note that the TC center position is shifted between VIIRS and GOES images due to parallax.

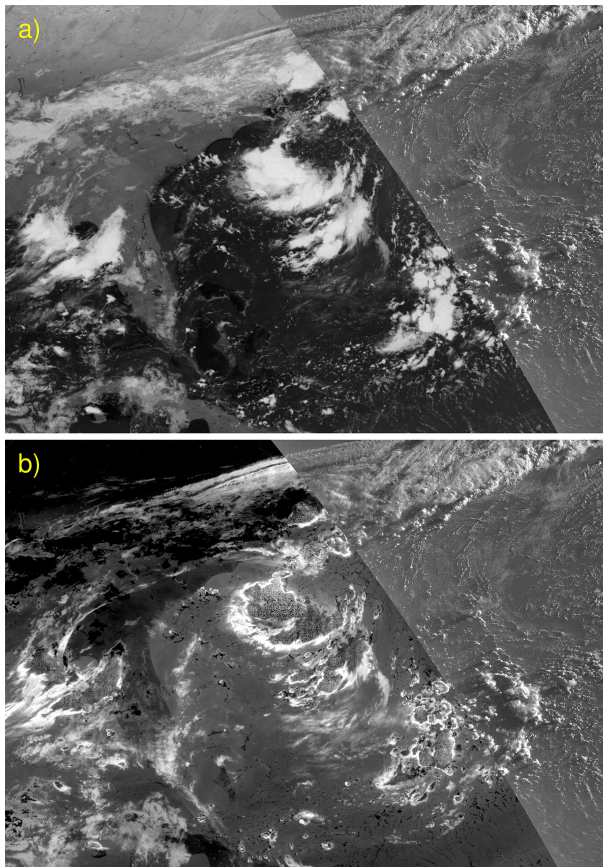


FIG. 12. Example of (a) ProxyVis use at NHC for the Atlantic Tropical Depression TD03 (2018) at 0956 UTC 7 Jul 2018 and (b) the fog product for the same scene. The fog product was operational at NHC before ProxyVis became available. The fog product is plotted using the same BT limits as in Fig. 2 and has also been normalized to match the range of values in the visible part of the image.

superior to ProxyVis, the advantage of ProxyVis is the similarity to visible imagery and the continuity across the terminator. ProxyVis is trained on VIIRS IR channels that have closely matching channels on all current and next-generation geostationary satellites, *GOES-16/17/18*, *Himawari-8/9*, *Meteosat-9/10/11*, *Geo-KOMPSAT-2A*, *MTG*, and *GeoXO*. GeoProxyVis applies ProxyVis to geostationary data so that it can be animated, and combines nighttime ProxyVis with daytime visible imagery normalized by solar zenith angle in a single product that has a relatively seamless transition at the day/night terminator. The true value of ProxyVis is that it can be applied to all current geostationary satellites to provide animated global GeoRing⁴ GeoProxyVis imagery. Figure 16 shows a comparison of visible ($0.64\ \mu\text{m}$) and GeoProxyVis for five geostationary satellites [available in real-time on SLIDER].

⁴ “GeoRing” refers to the global constellation of Geostationary Satellites positioned at different longitudes that form a kind of imaginary “ring” around Earth.

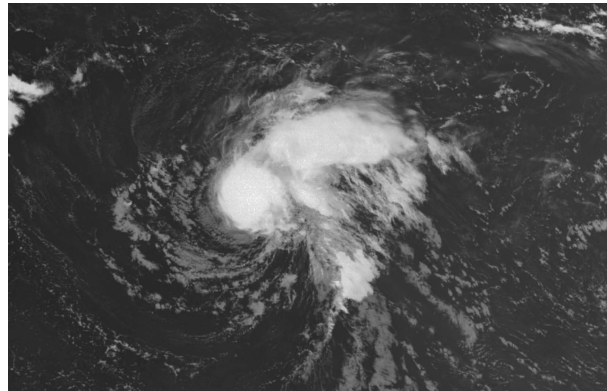


FIG. 13. Example of ProxyVis use at NHC showing then Tropical Storm Florence (2018) following a shear-induced period of dramatic weakening at 0800 UTC 7 Sep 2018 when operational intensity estimates were 55 kt.

Further, the simple, *single-channel* versions of ProxyVis enable its use for earlier GOES satellites that are still used in operations (e.g., *GOES-13* and *GOES-15*) but do not have all channels required for the main ProxyVis algorithm. The *single-regression* versions of ProxyVis can be used to create multiyear datasets of ProxyVis imagery that can be used in ML algorithms. ProxyVis demonstrates that it is possible to combine just a few commonly available IR channels using a simple tool, linear regression, to develop a nighttime proxy for visible imagery that significantly improves forecasters’ ability to track low-level oceanic clouds and circulation features at night, works for all geostationary satellites, and is useful across a wide range of backgrounds and meteorological scenarios.

Due to the simplicity of its formulation, ProxyVis is easy to implement in operations and has been used by NHC since 2018, by OPC and WPC since 2020, and by JTWC and some other NWS offices since 2022. Based on NHC and JTWC forecaster feedback, ProxyVis is a significant improvement for tracking low-level nighttime clouds compared to legacy products and can be used for multiple meteorological and TC applications, including determining the initial position for Dvorak intensity estimates and tracking tropical waves. ProxyVis is currently being assessed as a nighttime layer in GeoColor (Miller et al. 2020) and might be useful as a replacement for a visible band in other multispectral products (e.g., RGBs). Nonetheless, the current ProxyVis product has room for improvement. Limitations based on forecaster feedback are currently being used to guide the development of ProxyVis2, an enhanced ProxyVis product that has the goal to display nighttime oceanic low-level clouds similar to or better than ProxyVis, but also aims to closer replicate visible imagery in a variety of conditions, including over land and water, at high and low latitudes, and for multiple synoptic situations and meteorological phenomena. For ProxyVis2 we are using ML techniques and investigate the use of spatial patterns in the imagery (via convolutional neural networks) to improve the representation of different cloud types, which often have different spatial structures. Additional improvement might be provided by using geostationary

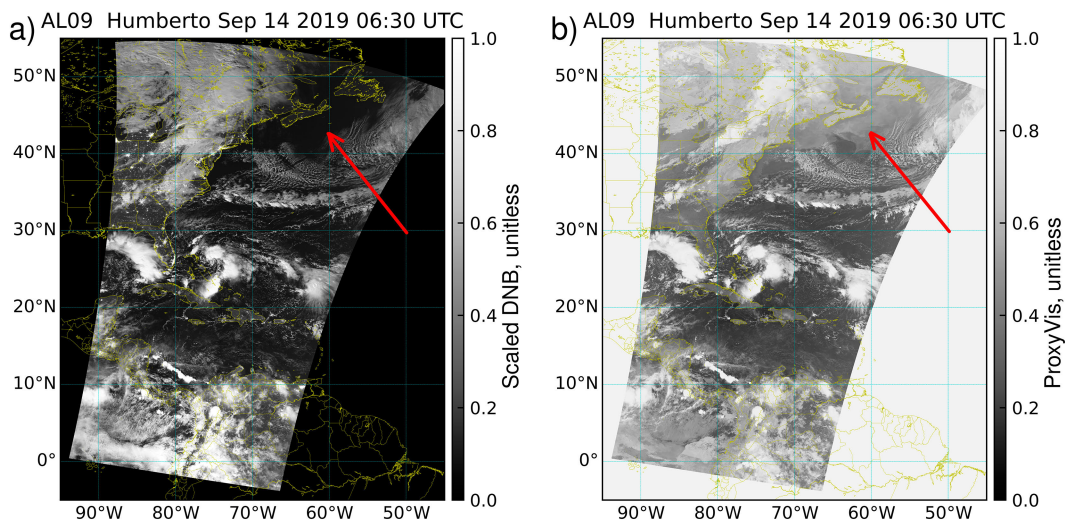


FIG. 14. Comparison of (a) VIIRS DNB and (b) PrVis for VIIRS data near the developing Atlantic Hurricane Humberto at 0630 UTC 14 Sep 2019. Red arrows indicate the region of sharp SST gradients where the ProxyVis image significantly differs from the DNB image.

satellite data for training, which would allow the use of additional channels but could also introduce errors due to varying viewing angles and thus differing parallax between VIIRS and GOES. Using a better formulation for the visible part and different ways of combining daytime and nighttime imagery is also investigated with ProxyVis2.

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Andrew Orrison, Buck Sampson, Curtis Seaman, Melinda Surratt, Natalie Tourville, Joe Zajic, and anonymous reviewers for their help with archiving data at CIRA, making ProxyVis imagery available to forecasters and providing suggestions for improvements. We would also like to thank NHC and JTWC forecasters for evaluating ProxyVis products and providing feedback. The scientific results and conclusions, as well as any views or opinions expressed herein, are those of the author(s) and do not necessarily reflect those of NOAA or the Department of Commerce.

Data availability statement. VIIRS and GOES-16/17 data used in the paper are publicly available from NOAA CLASS (2023).

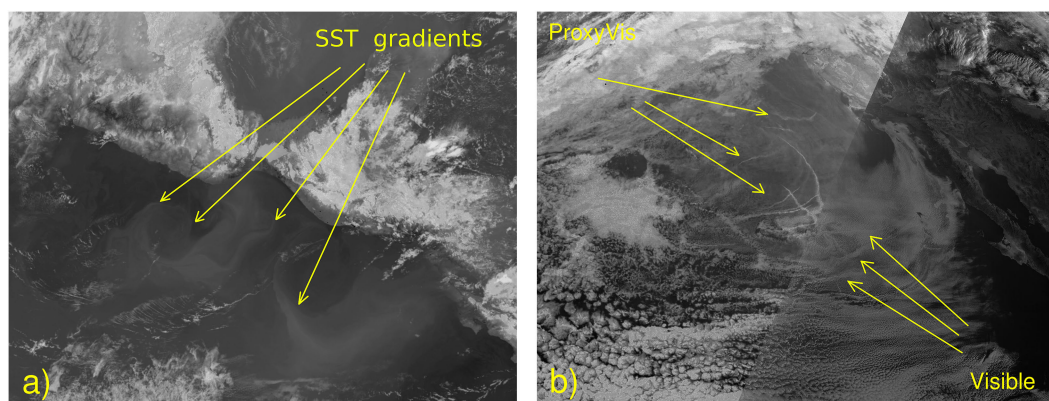


FIG. 15. (a) SST gradients in the eastern Pacific at 0130 UTC 17 Jan 2020, as seen in GOES-16 PrVis. The SST gradients, indicated by yellow arrows, can be easily distinguished from clouds on animated ProxyVis imagery. (b) Ship tracks in the eastern Pacific at 1500 UTC 13 Dec 2019, as seen in GOES-17 PrVis. The ship tracks, indicated by yellow arrows, are highlighted in ProxyVis similar to or better than in the visible ($0.64\ \mu\text{m}$) channel.

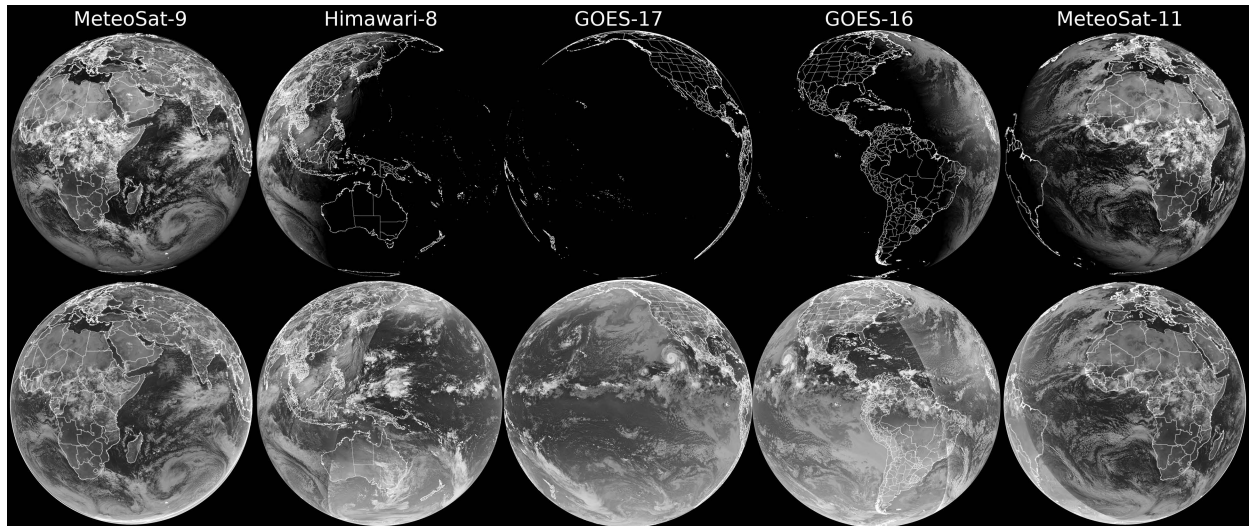


FIG. 16. Comparison of (top) visible ($0.64 \mu\text{m}$) and (bottom) GeoProxyVis for five operational (at the time the images were taken) geostationary satellites. The data for *Meteosat-9*, *Himawari-8*, *GOES-17*, *GOES-16*, and *Meteosat-11* are shown from left to right, respectively. Data for all satellites are at 0930 UTC 31 Jul 2022. Hurricane Frank (2022) can be seen in the eastern Pacific in *GOES-16/17* images.

APPENDIX A

Stretched $3.9\text{-}\mu\text{m}$ Imagery

While there is no single commonly accepted algorithm for making stretched $3.9\text{-}\mu\text{m}$ channel imagery, the general idea is to use a significant portion of the 255-color range to display a range of temperatures corresponding to low-level clouds, from about 270 to 330 K. For the examples shown in this paper, first, the bilinear scaling is applied to the $3.9\text{-}\mu\text{m}$ channel:

$$\text{ICn} = 255 \quad \text{for } T_b < 163.0, \quad (\text{A1})$$

$$\text{ICn} = 418 - T_b \quad \text{for } 163 \leq T_b < 242, \quad (\text{A2})$$

$$\text{ICn} = 660 - 2 \times T_b \quad \text{for } 242 \leq T_b < 330, \quad (\text{A3})$$

$$\text{ICn} = 0 \quad \text{for } T_b \geq 330.0, \quad (\text{A4})$$

where T_b is the $3.9\text{-}\mu\text{m}$ channel BT and ICn is the color number for the 8-bit (0–255) color table. Next, the resulting values are plotted using a color table from Table A1.

TABLE A1. Color table used for plotting bilinearly scaled $3.9\text{-}\mu\text{m}$ channel data to enhance low-level clouds.

Color No.		Blue		Green		Red	
Min	Max	Min	Max	Min	Max	Min	Max
0	1	230	0	230	0	230	0
2	40	0	0	0	0	0	0
41	80	2	90	2	90	2	90
81	123	91	153	91	153	91	153
124	255	154	255	154	255	154	255

APPENDIX B

Gamma DNB Scaling

The gamma-DNB scaling is an empirical algorithm based on the distribution of radiances within each scene. It implicitly accounts for the available moonlight, without using the moon phase information directly, and is designed to enhance cloud features at the expense of extreme minima (dark ocean) and extreme maxima (city/ship lights or moon glint). “Gamma DNB” has been evaluated on a large number of TC-centered scenes and proven to work well for a wide range of conditions around TCs, including different moon phases, and the presence of moon glint, city lights, and lightning. The scaling should be applied before the regridding step described in section 2c. The first step of the scaling process is the calculation of R_{skew} , the Fisher–Pearson coefficient of skewness of the original radiances in the scene. Next, to account for extremes in observed radiances, which often represent noncloud features (i.e., city lights), the 1st (R_{min}) and 99th (R_{max}) percentiles of the image scene are determined. The “inverse radiance ratio” (R_{minmax} , named similarly to the economic term “wage ratio”) is then determined as

$$R_{\text{minmax}} = \frac{\max(R_{\text{min}}, R_{\text{min_nozero}})}{R_{\text{max}}}, \quad (\text{B1})$$

where $R_{\text{min_nozero}}$ is the minimum nonzero value between R_{min} and R_{max} . The R_{min} is next adjusted to always be above zero to remove small negative radiances that sometimes appear because of the instrument calibration process (Geis et al. 2012; Lee et al. 2014). Next, radiances that have values above R_{max} are assigned R_{max} , and radiances below R_{min} are assigned R_{min} to obtain adjusted radiances R_{adj} . Using R_{adj} , R_{gamma} is calculated:

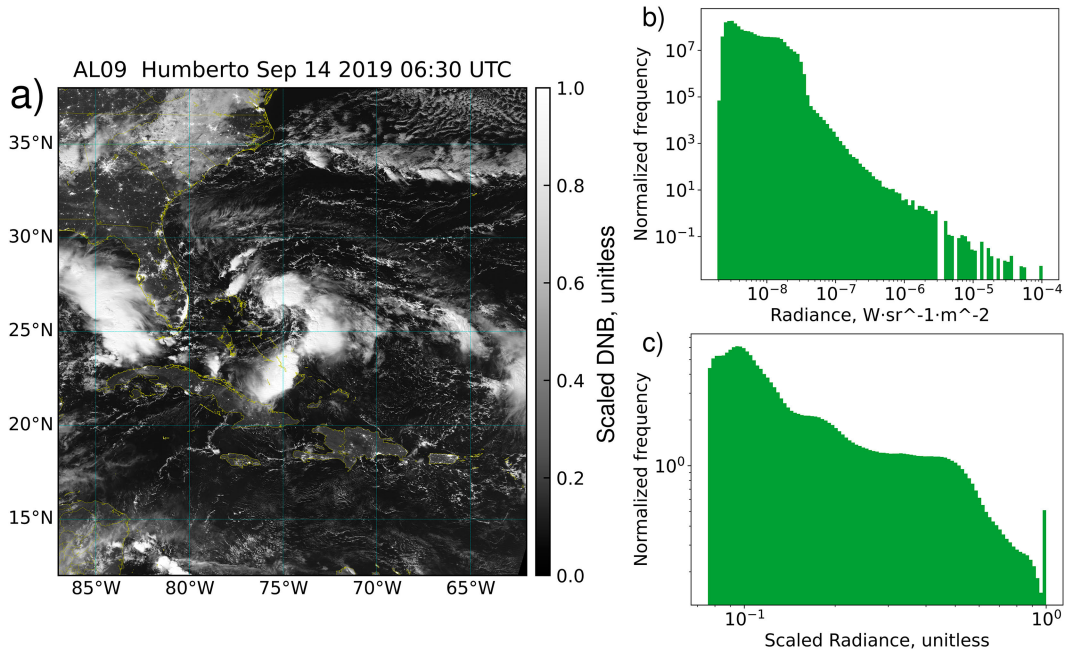


FIG. B1. An example of VIIRS DNB scaling described by (B1)–(B6). Shown are (a) gamma-scaled DNB for the Atlantic Hurricane Humberto (2019) at 0630 UTC 14 Sep 2019, one day after it formed to the southeast of Florida. The “raw” unscaled DNB image would look completely black if plotted using a linear color table. As an illustration, the histogram of values in the (b) raw and (c) scaled DNB images for this scene is shown. Note that the range of values in the raw image is from 10^{-9} to 10^{-4} while the scaled version normalizes values between 0 and 1 while preserving the general shape of the distribution. The x axis showing radiances is logarithmic in both (b) and (c). The time of the figure is during the full moon that occurred on 14 Sep 2019, which is the time of the best DNB performance.

$$R_{\text{gamma}} = R_{\text{adj}}^{1/3}. \quad (\text{B2})$$

This step is a gamma correction, which acts to brighten the images in radiance space. Next, R_{gamma} is normalized and gamma corrected based on the value of R_{skew} , as

$$R_{\text{scaled}} = \left[\frac{R_{\text{gamma}} - \min(R_{\text{gamma}})}{\max(R_{\text{gamma}}) - \min(R_{\text{gamma}})} \right]^{\gamma}, \quad (\text{B3})$$

where

$$\gamma = 1.6 \quad \text{for } R_{\text{skew}} > 10, \quad (\text{B4})$$

$$\gamma = 3 \quad \text{for } R_{\text{skew}} \leq 10. \quad (\text{B5})$$

The γ values from (B4) and (B5) act to reduce the number of the brightest pixels in the scene and make the distribution more linear and easier for the eye to interpret. If the inverse radiance ratio is large, indicating little variance in the scene (compared to the span of eight orders of magnitude in VIIRS DNB radiances), γ is further adjusted as

$$\gamma = 3 \quad \text{for } R_{\text{minmax}} > 0.02. \quad (\text{B6})$$

This allows for positive gamma expansion/dimming images with small radiance ranges, and gamma correction/brightening those with large radiance ranges.

An example of gamma–DNB scaling described by (B1)–(B6) is shown in Fig. B1 for the Atlantic Hurricane Humberto (2019), one day after it formed to the southeast of Florida. The raw DNB image would look completely black if plotted using a linear color table. Figures B1b and B1c show the histogram of values in the raw and scaled DNB images, respectively. Note that the range of values in the raw image is 10^{-9} – 10^{-4} $\text{W cm}^2 \text{sr}^{-1}$, and the scaled version normalizes values between 0 and 1 while preserving the general shape of the distribution. The time of the figure is during the full moon that occurred on 14 September 2019, representative of the time of the best DNB performance. For past (2014–17) and current examples, see RAMMB/CIRA (2019) and reprojected VIIRS DNB (2023).

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